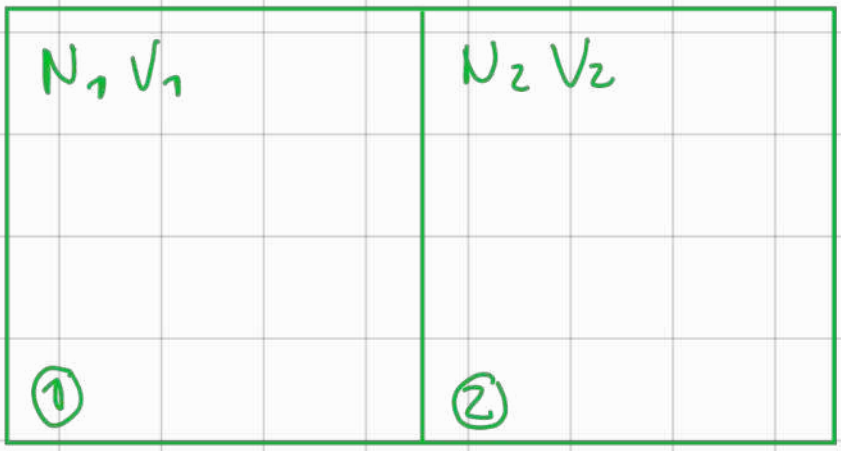
28/09/26

- RECALL OF STATISTICAL MECHANICS / NOTATION
- EQUILIBRATION
- MICROCANONICAL ENSEMBLE
- NTV ENSEMBLE

MICROCANONICAL ENSEMBLE



HYPOTHESES

- ISOLATED SYSTEM
- ① AND ② WEAKLY COUPLED
- MICROSCOPIC EQUIPROBABILITY

TOTAL ENERGY

$$E = E_1 + E_2 \Rightarrow E_2 = E - E_1$$

E IS CONSERVED

LET

$\Omega_1(E_1) = \{$ # MICROSCOPIC STATES OF SYSTEM ① WITH ENERGY $E_1\}$

$\Omega_2(E_2) = \{$ # MICROSCOPIC STATES OF SYSTEM ② WITH ENERGY $E_2\}$

TOTAL NUMBER OF MICROSCOPIC STATES

$$\Omega(E_1, E_2) = \Omega_1(E_1)\,\Omega_2(E_2)$$

Ω, Ω_1 AND Ω_2 ARE HUGE HENCE USE OF LOGARITHM CONVENIENT, I.E.

$$\ln[\Omega(E_1, E_2)] = \ln[\Omega_1(E_1)] + \ln[\Omega_2(E_2)]$$

BUT $E_2 = E - E_1$ SO THAT

$$\ln[\Omega(E_1, E-E_1)] = \ln[\Omega_1(E_1)] + \ln[\Omega_2(E-E_1)]$$

QUESTION:

WHAT IS THE MOST LIKELY VALUE OF E_1 ?

WE MUST ENFORCE THE CONDITION

$$\frac{\partial \ln[\Omega(E_1, E-E_1)]}{\partial E_1} = 0$$

I.E.



$$\frac{\partial \ln[\Omega_1(E_1)]}{\partial E_1} - \frac{\partial \ln[\Omega_2(E_2)]}{\partial E_2}\Big|_{E_2 = E - E_1} = 0$$

$$\frac{\partial \ln[\Omega_1(E_1)]}{\partial E_1} = \frac{\partial \ln[\Omega_2(E_2)]}{\partial E_2}\Big|_{E_2 = E - E_1}$$

IF WE DEFINE

$$\frac{\partial[\ln \Omega_1(E_1)]}{\partial E_1}\Big|_{N_1 V_1} = \beta_1$$

$$\frac{\partial \ln \Omega_2(E_2)}{\partial E_2}\Big|_{N_2 V_2} = \beta_2$$

9 WE HAVE THE CONDITION:

$$\beta_1 = \beta_2$$

WHAT DOES IT MEAN?

$$[\beta] = \frac{1}{[E]}$$

DEFINE

$$S \equiv k_B \ln[\Omega(N, V, E)] =$$

$$S(N, V, E)$$

AND WE CALL IT ENTROPY

WE HAVE THAT

$$\frac{\partial (k_B \ln \Omega_1(E_1))}{k_B \partial E_1} = \frac{\partial (k_B \ln \Omega_2(E_2))}{k_B \partial E_2}$$

$$\Downarrow$$

$$\frac{1}{k_B}\frac{\partial S_1}{\partial E_1} = \frac{1}{k_B}\frac{\partial S_2}{\partial E_2}$$

BUT $\left[\frac{\partial S}{\partial E_1}\right] = \frac{1}{[T]}$

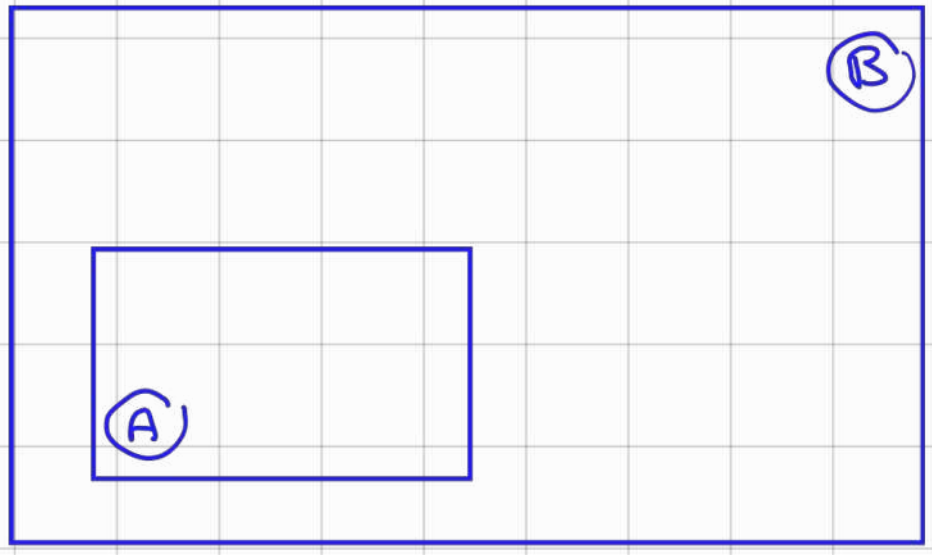
HENCE

$$\frac{1}{k_B T_1} = \frac{1}{k_B T_2}$$

I.E. $\boxed{T_1 = T_2}$ EQUILIBRIUM CONDITION.

AND $\beta_1 = \frac{1}{k_B T}$ $\beta_2 = \frac{1}{k_B T}$

CANONICAL ENSEMBLE



HYPOTHESES

- SYSTEM IS ISOLATED
- $E_A << E_B$ (I.E. (B) IS A THERMAL BATH)
- (A) AND (B) WEAKLY COUPLED

$|i\rangle$ QUANTUM STATE OF (A)

P_i = {PROBABILITY FOR (A) OF BEING IN STATE $|i\rangle$}

GOAL IS TO CALCULATE P_i

E_i = {ENERGY OF (A) BEING IN STATE $|i\rangle$}

$E_i + E_B = E$

$$\Omega(i, E_B) = \Omega(i, E - E_i)$$

$$= \Omega_A(i) \cdot \Omega_B(E - E_i)$$

AND WE CAN DEFINE THE FOLLOWING PROBABILITY

$$P_i \equiv \frac{\Omega(i, E_B)}{\sum_i \Omega(i, E_B)}$$

AGAIN WE MAKE USE OF LOGARITHM

$$\ln \Omega(i, E - E_i) =$$

$$\ln \Omega_A(i) + \ln \Omega_B(E - E_i)$$

BUT $E_i \ll E$ HENCE

$$\ln \Omega_B(E - E_i) \approx$$

$$\ln \Omega_B(E) - E_i \left.\frac{\partial \ln \Omega_B}{\partial E_B}\right|_{E_B = E - E_i} \equiv \beta$$

MOREOVER

$$\Omega_A(i) = 1 \Rightarrow \ln \Omega_A(i) = 0$$

THUS HAVING

$$\ln \Omega(i, E-E_i) \approx$$

$$\ln \Omega_B(E) - \beta E_i$$

BY PLUGGING THIS INTO P_i DEFINITION ONE HAS

$$P_i = \frac{\Omega_B(E)\, e^{-\beta E_i}}{\sum_i \Omega_B(E) \cdot e^{-\beta E_i}} =$$

$$= \frac{e^{-\beta E_i}}{\sum_i e^{-\beta E_i}}$$

IF WE DEFINE $Q = \sum_i e^{-\beta E_i}$

WE HAVE:

$$P_i = \frac{e^{-\beta E_i}}{Q}$$

Q IS CALLED THE

PARTITION FUNCTION

FROM Q ONE CAN CALCULATE THE INTERNAL ENERGY U:

$$U \equiv \langle E \rangle = \frac{1}{Q} \sum_i E_i e^{-\beta E_i} =$$

$$Q = \sum_i e^{-\beta E_i}$$

BUT $$\frac{\partial \ln Q}{\partial \beta} = \frac{\partial}{\partial \beta}\left(\sum_i e^{-\beta E_i}\right) =$$

$$= -\frac{1}{Q}\sum_i E_i e^{-\beta E_i} = -U$$

HENCE

$$U = -\frac{\partial \ln Q}{\partial \beta}$$

Q. IS THERE A RELATION BETWEEN FREE ENERGY AND Q ?

FROM THERMODYNAMIC:

$$F = U - TS \qquad F = F(T, V)$$

$$\frac{\partial (F/T)}{\partial (1/T)}\bigg|_V = \frac{1}{T}\frac{\partial F}{\partial (1/T)} + F =$$

$$= \frac{1}{T}\frac{\partial F}{\partial T}\bigg|_V \frac{\partial T}{\partial (1/T)} + F =$$

$$= \frac{1}{T} \left.\frac{\partial F}{\partial T}\right|_V \frac{1}{\frac{\partial (1/T)}{\partial T}} + F =$$

$$= -T \left.\frac{\partial F}{\partial T}\right|_V + F = \left.\frac{\partial (F/T)}{\partial (1/T)}\right|_V$$

BUT $\left.\frac{\partial F}{\partial T}\right|_V = -S$ HENCE

$$= TS + F = TS + U - TS$$

$$= U \qquad \text{I.G.}$$

$$\frac{\partial (F/T)}{\partial (1/T)} = U = -\frac{\partial (\ln Q)}{\partial \beta} = \frac{\partial (\ln Q)}{\partial \left(\frac{1}{k_B T}\right)}$$

FROM WHICH FOLLOWS THAT

$Q = \sum_i e^{-\beta E_i}$

$$\frac{F}{T} = -k_B \ln Q$$

$$\boxed{F = -k_B T \ln Q}$$

WORKHORSE OF STATISTICAL MECHANICS

CLASSICAL LIMIT OF STATISTICAL MECHANICS

[$\hbar \to 0$]

LET $\hat{A}$ BE AN OBSERVABLE

$$\langle \hat{A} \rangle = \frac{\sum_i \langle i|\hat{A}|i\rangle e^{-\beta E_i}}{Q} \qquad p_i = \frac{e^{-\beta E_i}}{Q}$$

↑ STATISTICAL AVERAGE

WHERE $\hat{\mathcal{H}}|i\rangle = E_i|i\rangle$ ← AUTOSTATES OF $\hat{\mathcal{H}}$

INSTEAD OF EIGENSTATES OF $\hat{\mathcal{H}}$ WE CAN USE A GENERIC COMPLETE SET, I.E.

$$\langle \hat{A} \rangle = \frac{\sum_{\alpha'\alpha} \langle \alpha | \hat{A} e^{-\beta \hat{\mathcal{H}}} | \alpha \rangle}{Q}$$

WHERE $|\alpha\rangle$ IS A COMPLETE (*) SET OF STATES, I.E.

$$\sum_\alpha |\alpha\rangle\langle\alpha| = \mathbb{1}$$

↑ IDENTITY

↑ ARBITRARY! NON DIAGONAL TERMS ARE 0 → POSTULATE (SEE HUANG 8.1)

LET'S ASSUME FURTHER THAT

$$\hat{A} = \hat{A}(\hat{q}) \qquad \hat{\mathcal{H}} = \hat{K}(\hat{p}) + \hat{U}(\hat{q})$$

WITH $\hat{K} = \sum_{i}^{N} \frac{\hat{p}_i^2}{2m}$ $\qquad \hat{U} = \sum_i^N \frac{\kappa}{2} \hat{q}_i^2$

(*) NOT NECESSARILY ORTHONORMAL

$$\int d^{3N}q \, |q\rangle\langle q| = \mathbb{1}$$

$$\int d^{3N}p \, |p\rangle\langle p| = \mathbb{1}$$

SO THAT

$$\langle \hat{A} \rangle = \frac{\int d^{3N}q \, \langle q | \hat{A} e^{-\beta \hat{\mathcal{H}}} | q \rangle}{\mathcal{Q}}$$

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

YOU CAN EASILY PROVE THAT:

$$[\hat{A}, \hat{B}\hat{C}] = \hat{B}[\hat{A}, \hat{C}] + [\hat{A}, \hat{B}]\hat{C} \quad (1)$$

AND

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B} \quad (2)$$

FROM THESE, IF $F(\hat{p})$ AND $G(\hat{q})$ ARE SMOOTH FUNCTIONS OF $\hat{p}$ AND $\hat{q}$ RESPECTIVELY, ONE HAS

$$[\hat{q}, F(\hat{p})] = i\hbar \frac{\partial F(\hat{p})}{\partial \hat{p}} \quad (3)$$

AND

$$[\hat{p}, G(\hat{q})] = -i\hbar \frac{\partial G(\hat{q})}{\partial \hat{q}} \quad (4)$$

[SEE SAKURAI]
SEC. 2.2 PAG. 85

WE HAVE:

$$-\beta \hat{\mathcal{H}} = -\beta \hat{K} - \beta \hat{V}$$

BCH FORMULA TELLS US

$$e^{\hat{A}} e^{\hat{B}} = e^{\hat{A}+\hat{B}+\frac{1}{2}[\hat{A},\hat{B}]+\frac{1}{12}([\hat{A},[\hat{A},\hat{B}]]+\ldots}$$

$$\ldots + [\hat{B},[\hat{B},\hat{A}]]) + \cdots$$

$$\hat{A}+\hat{B} = -\beta \hat{\mathcal{H}}$$

IF WE SET $\hat{A} = -\beta \hat{K}$ AND

$\hat{B} = -\beta \hat{V}$ BCH FORMULA YIELDS:

$$e^{-\beta \hat{K}} e^{-\beta \hat{V}} \approx e^{-\beta(\hat{K}+\hat{V}+\frac{1}{2}[\hat{K},\hat{V}]+\ldots}$$

$$\ldots + \frac{1}{12}([\hat{K},[\hat{K},\hat{V}]] + [\hat{V},[\hat{V},\hat{K}]]))$$

BUT $[\hat{K},\hat{V}] = \frac{k}{4m}[\hat{p}^2,\hat{q}^2] =$

$$= -\frac{k}{4m} 2i(\hat{q}\hat{p}+\hat{p}\hat{q})k = \mathcal{O}(\hbar)$$

AND SIMILARLY

$$[\hat{K},[\hat{K},\hat{V}]] = O(\hbar^2)$$
$$[\hat{V},[\hat{V},\hat{K}]] = O(\hbar^2)$$

WHERE WE MADE USE OF RELATIONS (1) - (4)

HENCE IN THE LIMIT OF $\hbar \to 0$ WE CAN NEGLECT $O(\hbar)$ TERMS

NOTE THAT $\hbar \to 0$ IS EQUIVALENT TO LET $\Lambda = \sqrt{\frac{2\pi\hbar^2}{m k_B T}} \to 0$

SO THAT

$$e^{-\beta\hat{\mathcal{H}}} = e^{-\beta\hat{K}-\beta\hat{V}} \approx e^{-\beta\hat{K}} e^{-\beta\hat{V}}$$

LET'S WORKOUT THE PARTITION FUNCTION

$$Q = \int d^{3N}q \langle q | e^{-\beta\hat{\mathcal{H}}} | q \rangle =$$

$$= \int d^{3N}q \langle q | e^{-\beta\hat{K}} e^{-\beta\hat{V}} | q \rangle$$

IF WE INSERT $\int d^{3N}p \, |p\rangle\langle p| = \mathbb{I}$

WE HAVE

$$= \int d^{3N}q \, d^{3N}p \, \langle q|e^{-\beta\hat{K}}|p\rangle\langle p|e^{-\beta\hat{V}}|q\rangle$$

FUNCTIONS OF q AND p!

$$= \int d^{3N}q \, d^{3N}p \; e^{-\beta(K(p)+V(q))} \langle q|p\rangle\langle p|q\rangle$$

BUT $$\langle q|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{i\frac{p\cdot q}{\hbar}}$$

$$\langle p|q\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{-i\frac{q\cdot p}{\hbar}}$$

[SEE SAKURAI SEC. 1.7 PAG. 54-56]

HENCE WE OBTAIN

$$\langle q|p\rangle\langle p|q\rangle = \frac{1}{2\pi\hbar}$$

THUS HAVING

FUNCTION OF q AND p

$$Q = \int \frac{d^{3N}q \, d^{3N}p}{(2\pi\hbar)^{3N} N!} e^{-\beta \mathcal{H}(q,p)}$$

CORRECT BOLTZMANN WAVE FUNCTIONS MUST BE EITHER SYM. OR ANTISYM.

SIMILARLY ONE CAN TREAT

$$\int d^{3N}q \, \langle q | \hat{A} e^{-\beta \hat{\mathcal{H}}} | q \rangle =$$

TO HAVE

$$= \int \frac{d^{3N}q \, d^{3N}p}{(2\pi\hbar)^{3N} N!} A(q,p) \, e^{-\beta \mathcal{H}(q,p)}$$

$\mathcal{H} = K + V$

TO FINALLY HAVE IN THE LIMIT OF $\hbar \to 0$

$$\langle A \rangle = \frac{\int \frac{d^{3N}q \, d^{3N}p}{(2\pi\hbar)^{3N} N!} A \, e^{-\beta \mathcal{H}}}{\int \frac{d^{3N}q \, d^{3N}p}{(2\pi\hbar)^{3N} N!} e^{-\beta \mathcal{H}}} \quad Q$$

BUT SINCE $A = A(q)$, AS WE ASSUMED, ONE HAS

$$\langle A \rangle = \frac{\int A(q) e^{-\beta V(q)} d^{3N}q}{Z}$$

WHERE

Z IS THE CONFIGURATIONAL

PARTITION FUNCTION

$$Z = \int d^{3N}_{q} \; e^{-\beta V(q)}$$

ALTERNATIVE WAY

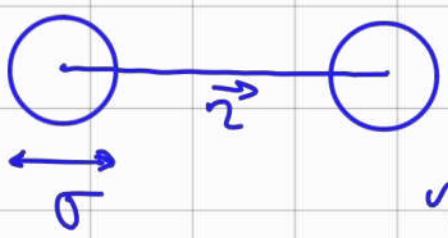
$\beta \to 0$ (HIGH TEMPERATURES)

JUST NOTE THAT

$$e^{-\beta(\hat{K}+\hat{V})} = \mathbb{1} + \beta(\hat{K}+\hat{V}) + \mathcal{O}(\beta^2) \approx \mathbb{1} + \beta(\hat{K}+\hat{V})$$

$$= e^{-\beta\hat{K}} \, e^{-\beta\hat{V}} + \mathcal{O}(\beta^2)$$

HARD SPHERE MODEL



$$r = |\vec{r}|$$

$$U_{HS}(r) = \begin{cases} \infty & r \le \sigma \\ 0 & r > \sigma \end{cases}$$

$$\phi = \{\text{VOLUME FRACTION}\} = \hat{v}_{HS}\frac{N}{V}$$

N: NUMBER OF PARTICLES; V: VOLUME

WHERE

$$\hat{v}_{HS} = \frac{\pi}{6}\sigma^3$$

$\hat{v}_{HS} N$ IS THE VOLUME OCCUPIED BY N SPHERES

Q. WHY HARD SPHERES CAN FORM A CRYSTAL?

HARD SPHERES ARE AN "ENTROPIC" SYSTEM SINCE

$$F = U - TS \quad \text{BUT} \quad U = \frac{3}{2}k_B T N + \cancel{U_{EX}}^{\,0}$$

U_{EX}: EXCESS ENERGY

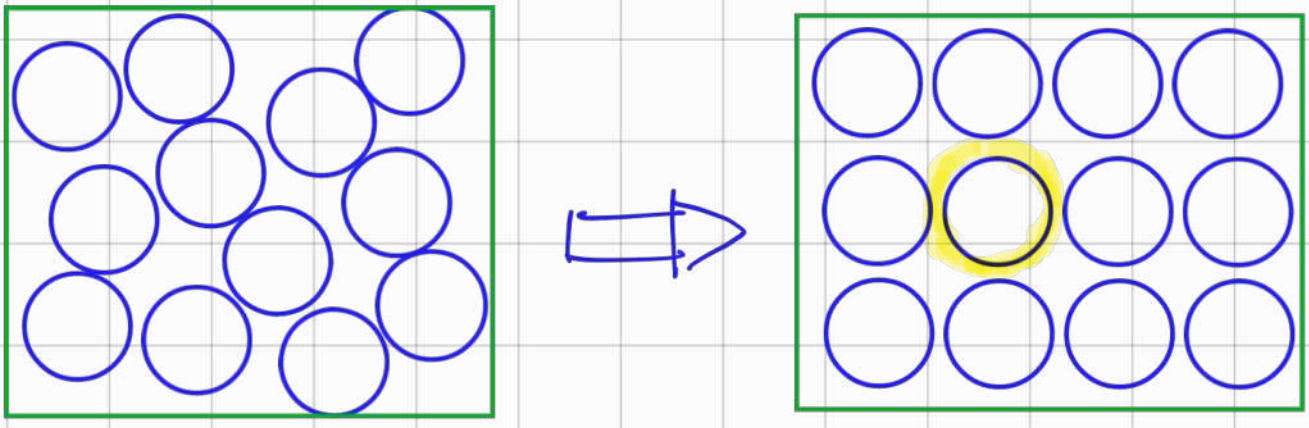
HENCE

$$F = \frac{3}{2}k_B T N - TS =$$

$$= T\left(\frac{3}{2}k_B N - S\right)$$

THIS IMPLIES THAT MINIMIZING F IS EQUIVALENT TO MAXIMIZE S (ENTROPY) I.E. THE NUMBER OF ACCESSIBLE STATES

BUT



↑ ALMOST ALL PARTICLES BLOCKED

GAIN OF ACCESSIBLE STATE, I.E. LARGER ENTROPY

HIT AND MISS MONTE CARLO: EXCLUDED VOLUME

IN 1D

$$\int_a^b f(x)\,dx = (b-a)\underbrace{\left(\frac{1}{b-a}\int_a^b f(x)\,dx\right)}_{\langle f\rangle} =$$

$$= (b-a)\langle f\rangle$$

IN 3 DIMENSIONS

$$\int_V f(\vec{x})\, d^3x = V\left(\frac{1}{V}\int f(\vec{x})\, d^3x\right) = V\langle f\rangle$$

VIRIAL EXPANSION

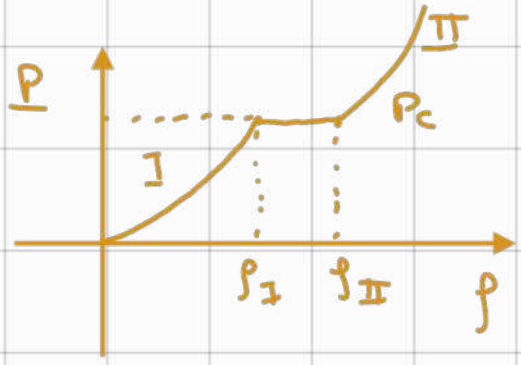
$$\beta P = \rho - \frac{1}{2}\rho^2 \int \left(e^{-\beta U_{HS}(r)} - 1\right) d^3r_2 + O(\rho^3)$$

← MAYER f-FUNCTION

LET $f = e^{-\beta U_{HS}(r)} - 1$

$$\beta P = \rho - \frac{1}{2}\rho^2 \int f(r)\, d^3r_2$$

IN PRESENCE OF A FIRST ORDER TRANSITION

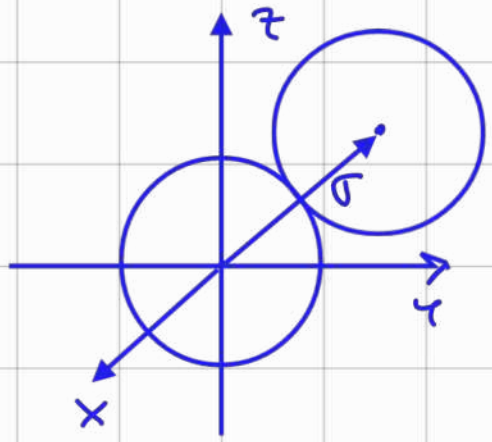


TWO PHASES I AND II (E.G. I = GAS AND II = LIQUID)

$$f = \begin{cases} -1 & r \le \sigma \\ 0 & r > \sigma \end{cases}$$

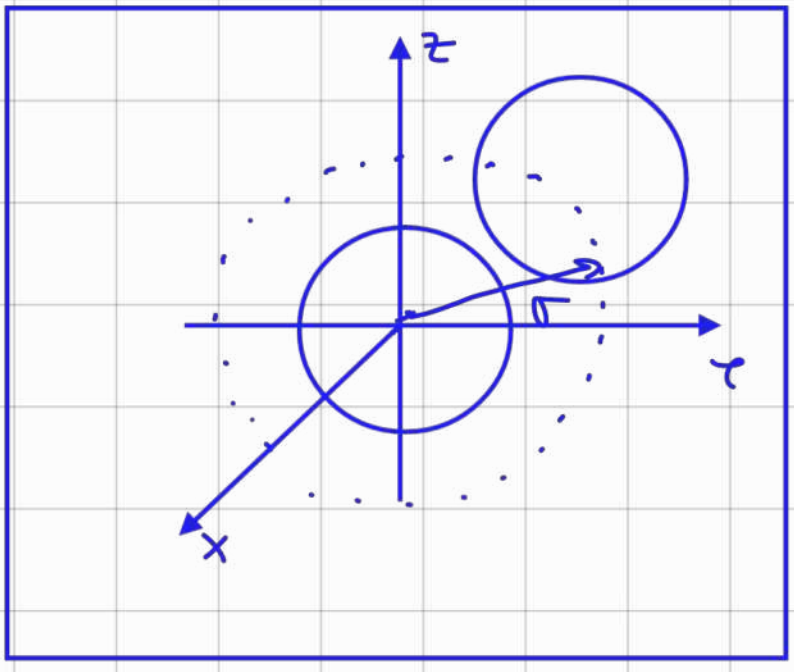
HENCE

$$-\int f(r)\, d^3r = \{\text{VOLUME OF A SPHERE OF RADIUS } \sigma\} = \underbrace{\frac{4\pi}{3}\sigma^3}_{v_{excl}}$$



$$\beta P = \rho + \frac{\rho^2}{2} U_{EXCL} + O(\rho^2)$$

WE CAN CALCULATE U_{EXCL} THROUGH A "HIT AND MISS" MONTE CARLO



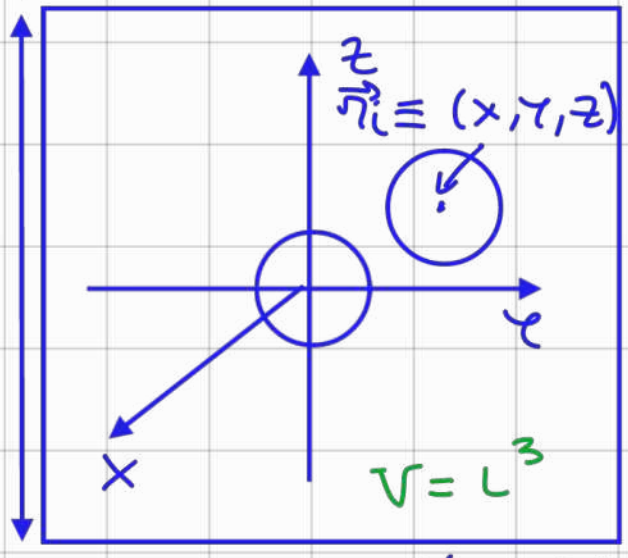
← BOX OF VOLUME V

$$U_{EXCL} = V\left(\frac{1}{V}\int_V \left(1 - e^{-\beta U_{MS}(r)}\right) d^3r\right)$$

$$= V \langle \tilde{f} \rangle \quad \text{WHERE} \quad \tilde{f} = -f$$

$\langle \tilde{f} \rangle$ CAN BE CALCULATED BY A "HIT AND MISS" MONTE CARLO BY INSERTING RANDOMLY A HS GIVEN ONE FIXED AT ORIGIN I.E.

$L^3 = V$



L

$x = \xi_x \, L/2$

$y = \xi_y \, L/2$

$z = \xi_z \, L/2$

ξ_α RANDOM NUMBER

$-1 \leq \xi_\alpha \leq 1$

$\alpha \in \{x, y, z\}$

$$\langle \tilde{f} \rangle = \frac{1}{N} \sum_i^N \tilde{f}(\vec{r}_i) = \frac{N_{OV}}{N}$$

$N = \{$ NUMBER OF INSERTIONS $\}$

$N_{OV} = \{$ NUMBER OF OVERLAPS $\}$

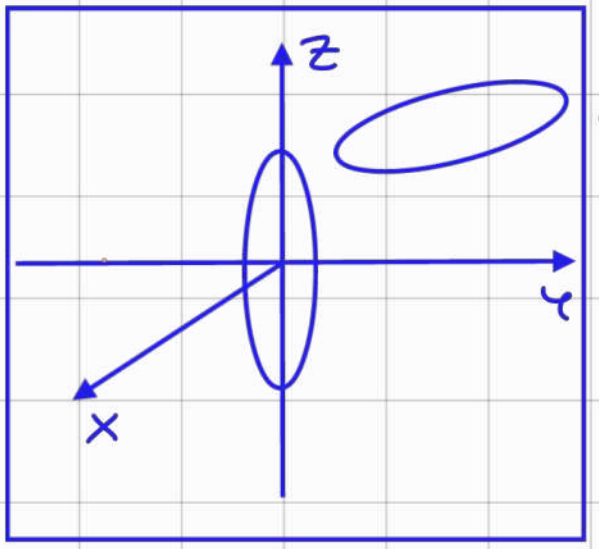
AND NOTE THAT

$$\tilde{f} = \begin{cases} 1 & r \leq \sigma \\ 0 & r > \sigma \end{cases}$$

$$\text{ERROR} \propto \frac{1}{\sqrt{N}}$$

$$V_{EXCL} = V \cdot \langle \tilde{f} \rangle = \frac{N_{OV}}{N}$$

MORE GENERALLY CONSIDER FOR EXAMPLE HARD ELLIPSOIDS (HE)



← BOX OF VOLUME V

$$U_{HE} = U_{HE}(\vec{r}, \Omega)$$

$$U_{EXCL} = \int \left(1 - e^{-\beta U_{HE}(\vec{r},\Omega)}\right) d^3r \frac{d\Omega}{4\pi} =$$

$$V \cdot \int \left(1 - e^{-\beta U_{HE}(\vec{r},\Omega)}\right) \frac{d^3r}{V} \frac{d\Omega}{4\pi} =$$

$$= V \cdot \langle \tilde{f} \rangle$$

$$\tilde{f} = \int \left(1 - e^{-\beta U_{HE}(\vec{r},\Omega)}\right) \frac{d^3r}{V} \frac{d\Omega}{4\pi}$$

INSERTION OF HARD ELLIPSOIDS WITH RANDOM POSITIONS AND ORIENTATIONS

$$\langle \tilde{f} \rangle = \frac{N_{OV}}{N}$$

N_{OV} ← NUMBER OF OVERLAPS

N ← NUMBER OF INSERTIONS

$$U_{EXCL} = V \langle \tilde{f} \rangle$$

SO THAT WE WILL HAVE A ROUGH ESTIMATE OF THEIR EQUATION OF STATE

$$\beta P = \rho + \frac{\rho^2}{2} U_{EXCL}$$

COMPLEX SYSTEMS

(ENSEMBLE OF N PARTICLES)

$A = A(q^N, p^N)$ $\quad q^N = \{\vec{q}_1 \dots \vec{q}_N\}$

$p^N = \{\vec{p}_1 \dots \vec{p}_N\}$

$$\langle A \rangle = \frac{1}{Q} \int \frac{d^{3N}q \, d^{3N}p}{(2\pi\hbar)^{3N} N!} A(q^N p^N) e^{-\beta \mathcal{H}}$$

THERMODYNAMIC AVERAGE

$$\mathcal{H} = K + U \qquad K = \sum_i \frac{\vec{p}_i^{\,2}}{2m}$$

$$U = U(\vec{q}_1 \dots \vec{q}_N) = \frac{1}{2} \sum_{i,j} v(\vec{q}_i - \vec{q}_j)$$

$$Q = \int \frac{d^{3N}q \, d^{3N}p}{(2\pi\hbar)^{3N} N!} e^{-\beta \mathcal{H}}$$

LET'S ASSUME $A = A(q^N)$

INTEGRATE OVER p^N SO THAT

$$\langle A \rangle = \frac{\int A(q^N) e^{-\beta U} d^{3N}q}{Z}$$

$$Z = \int e^{-\beta U} d^{3N}q$$

IF WE DEFINE

$$\mathcal{N}(q^N) \equiv \frac{e^{-\beta U(q^N)}}{Z}$$ ONE HAS

$$\langle A \rangle = \int_V \mathcal{N}(q^N)\, A(q^N)\, d^{3N}q$$

IF WE USE A STANDARD QUADRATURE (E.G. TRAPEZOIDAL INTEGRATION)

- WE HAVE $3N$ DIMENSIONS
- ASSUME 10 POINTS ALONG EACH DIRECTION
- ASSUME $N = 1000$ PARTICLES

WE WILL HAVE TO DO

10^{3N} OPERATIONS

IF OUR COMPUTER CAN DO EXAFLOPS, I.E. 10^{18} FLOPS (VERY POWERFUL!)

IT WILL TAKE

$$\frac{10^{3\cdot 1000}}{10^{18}} = 10^{3000-18} \approx 10^{3000}\ \text{s}$$

TO BE COMPARED WITH HUMAN LIFE

$3600 \times 24 \times 365 \times 100 \approx 10^9$ s

UNIVERSE? 10^{18} s ($\approx$ 20 BILLION YEARS)

YET NOT ENOUGH!

BUT MC CAN DO THE JOB!

$$\langle A \rangle = \int A\, \mathcal{N}(q^N)\, d^{3N}q =$$

(ENSEMBLE AVERAGE)

$$= V^N \left(\frac{1}{V^N} A\, \mathcal{N}(q^N)\, d^{3N}q \right) =$$

$$= V^N \cdot \langle A\, \mathcal{N} \rangle_V$$

← AVERAGE OF $A\mathcal{N}$ WHEN WE GENERATE RANDOMLY POINTS INSIDE VOLUME V

IN PRINCIPLE IS OK

BUT IT IS INEFFICIENT SINCE

IT IS VERY LIKELY THAT

$\mathcal{N}(q^N) \approx 0$ IF q^N IS RANDOM

NEED OF SOME IMPROVEMENT!

IMPORTANCE SAMPLING

1D CASE

$$I = \int_a^b f(x)\,dx = \int \left(\frac{f(x)}{w(x)}\right) w(x)\,dx = \left\langle \frac{f}{w} \right\rangle_w$$

WHERE $\int_a^b w\,dx = 1$

$$U(x) = \int_0^x w(x)\,dx'$$

BE $U = U(x)$

$$dU = w\,dx \qquad U(b) = 1 \quad U(a) = 0$$

$$= \int_0^1 \frac{f(x(U))}{w(x(U))}\,dU = \left\langle \frac{f}{w} \right\rangle_w$$

UNIFORM DISTRIBUTION

$$I = \frac{1}{L} \sum_i^L \frac{f(x(U_i))}{w(x(U_i))}$$

L IS THE NUMBER OF RANDOM POINTS GENERATED

IF $g_i \equiv \frac{1}{L}\frac{f_i}{w_i}$ WHERE $f_i = f(x(U_i))$, $w_i = w(x(U_i))$

$$\sigma^2(I) = \sigma^2\left(\sum_{i=1}^L g_i\right) =$$

$$= L\,\sigma^2(g_i) = L\,\frac{1}{L^2}\,\sigma^2\left(\frac{f}{w}\right)$$

I.E.

$$\sigma^2(I) = \frac{1}{\sqrt{L}}\,\sigma\left(\frac{f}{w}\right)$$

IF $\frac{f}{w} \simeq$ CONSTANT $\quad \sigma\left(\frac{f}{w}\right) \simeq 0$

LET'S APPLY THIS TECHNIQUE TO STATISTICAL AVERAGES $\omega = \omega(q_1 \dots q_N)$

$$\langle A \rangle = \int A(q^N) \frac{e^{-\beta U(q^N)}}{Z} d^{3N}q$$

(STATISTICAL AVERAGE)

$$= \left\langle \frac{AN}{\omega} \right\rangle_\omega$$

(f)

WISE CHOICE IS $\omega = N(q^N)$

HENCE

$$= \left\langle \frac{A\cancel{N}}{\cancel{N}_{=\omega}} \right\rangle_N = \frac{1}{L} \sum_i^L A(\vec{q}_1^{\,i} \dots \vec{q}_N^{\,i})$$

↑ UNWEIGHTED AVERAGE!

NOW WE NEED TO FIND A STRATEGY TO GENERATE POINTS ACCORDING TO DISTRIBUTION N

SINCE $N = \frac{e^{-\beta U}}{Z}$ BUT

Z IS AN INTEGRAL AND WE CAN'T CALCULATE IT EFFICIENTLY

MARKOV CHAIN

SET OF STOCHASTIC VARIABLES

$X_t \in \{$ FINITE SET OF POSSIBLE VALUES $\}$

1) $X_t = (\vec{q}_1 \dots \vec{q}_N)$ SET OF COORDS THAT ON A PC CAN ASSUME VALUES FROM A FINITE SET

2) $P(X_{t+1} = y_{t+1} \mid X_t = y_t)$ ← STATE OF THE SYSTEM (E.G. POSITIONS OF ALL PARTICLES)

3) CHAIN IS ERGODIC

NOTE THAT:

$$P(X_{t+1} = y_n \mid X_t = y_o) = \Pi_{on}$$

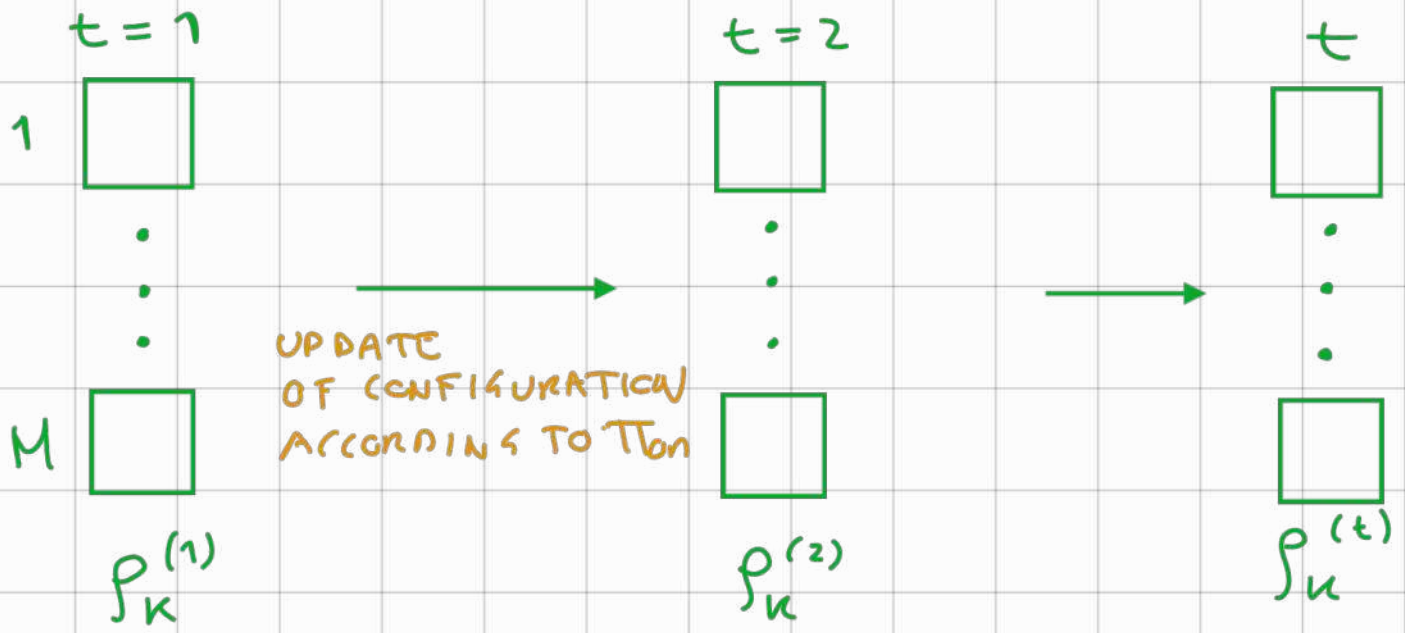
TRANSITION PROBABLITY ← TO GO FROM OLD STATE (o) TO NEW STATE (n)

CONSIDER M SYSTEMS WHICH WE ARE SIMULATING IN PARALLEL

$y_t = (\vec{x}_1^{(t)} \dots \vec{x}_N^{(t)})$ ← POSITIONS OF N PARTICLES (STATE OF THE SYSTEM)

NOTE THATE THEY CONSTITUTE A FINITE SET

HENCE WE CAN USE AN INDEX k TO IDENTIFY THEM



$\rho_k^{(t)}$ IS THE PROBABILITY OF HAVING STATE k

AT $t=1$ WE WILL BE GENERALLY "OUT OF EQUILIBRIUM"

BUT IF THE MARKOV CHAIN IS ERGODIC

$$\lim_{t \to +\infty} \rho_i^{(t)} = \rho_i$$

SYSTEM "EQUILIBRATES" AND TENDS TO EQUILIBRIUM DISTRIBUTION ρ_i

IN OTHER WORDS WHEN $\rho_i^{(t)} = \rho_i$ WE CAN SAY THAT SYSTEM IS EQUILIBRATED

NOTE THAT TRANSITION MATRIX IS THE SAME FOR ALL SYSTEMS

AND WE UPDATE SYSTEM'S STATE AS DISCUSSED IN THE FOLLOWING.

UPDATE OF SYSTEM'S STATE

CONSIDER A SYSTEM IN STATE i

TO UPDATE ITS STATE

$$\Pi = \begin{pmatrix} \vdots \\ \Pi_{i0} \dots \Pi_{ik} \dots \Pi_{iN_S} \\ \vdots \end{pmatrix} \leftarrow i\text{-th ROW}$$

PROBABILITY TO GO FROM STATE i TO STATE k

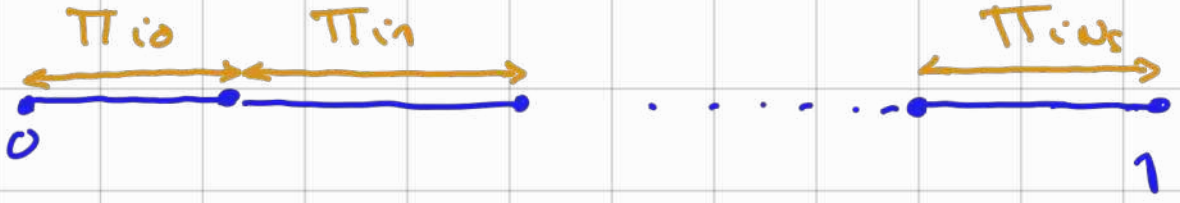
TRANSITION MATRIX

$N_S = \{$ TOTAL NUMBER OF POSSIBLE STATES $\}$

NOTE THAT $\sum_k \Pi_{ik} = 1$

GENERATE A UNIFORM

RANDOM NUMBER $\xi \in [0,1]$



IF $\xi < \Pi_{i0}$ THEN $i \to 0$

IF $\Pi_{i0} < \xi < \Pi_{i0} + \Pi_{i1}$ THEN $i \to 1$

$\vdots$

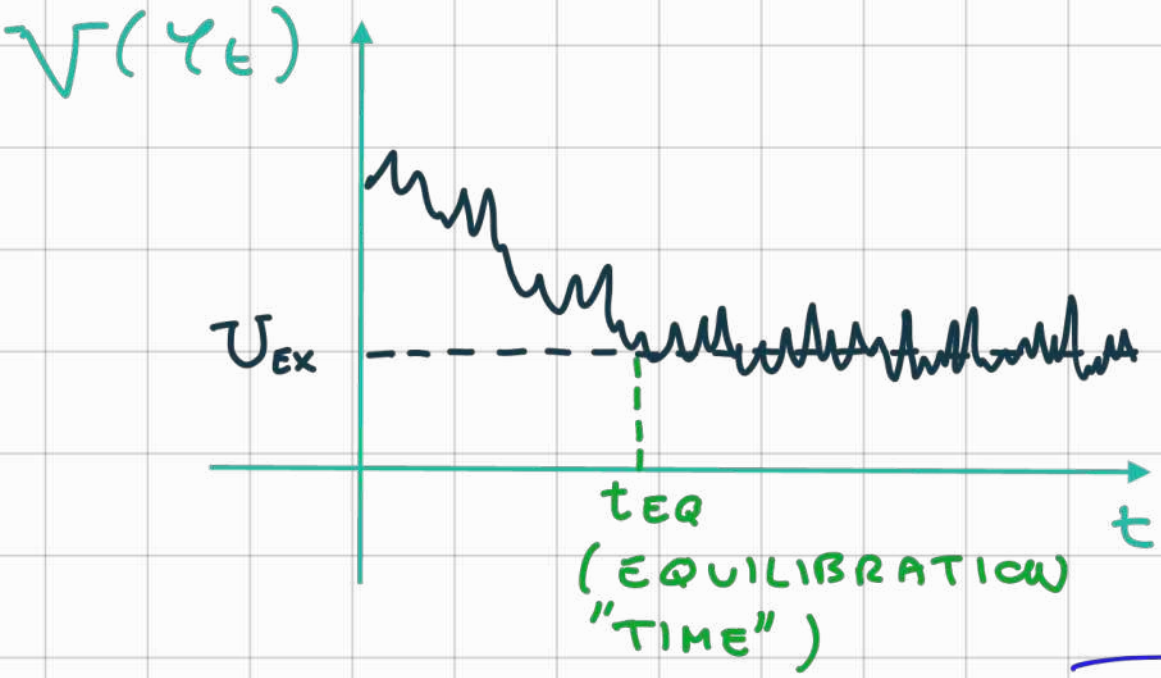
WE CAN VIEW "EQUILIBRATION" FROM A DIFFERENT PERPESCTIVE

LET'S CONSIDER THE EXCESS ENERGY DUE TO PARTICLES INTERACTIONS

$U = \{\text{INTERNAL ENERGY}\} = K + U_{EX}$

WHERE K IS THE KINETIC ENERGY AND IF $V = V(\vec{x}_1 \dots \vec{x}_N)$ IS THE POTENTIAL ENERGY OF THE SYSTEM

$$U_{EX} = \langle V(\vec{x}_1 \dots \vec{x}_N) \rangle$$



LET'S CONSIDER AGAIN THE M SYSTEM EVOLVING (RUNNING) IN PARALLEL

$\rho_n^{(t)} \cdot M = \{\#$ OF SYSTEM IN STATE $K\}$

$\rho_n^{(t+1)}$ AND $\rho_n^{(t)}$ CAN BE RELATED THROUGH Π_{kn}

INDEED:

$$\rho_n^{(t+1)} M = \rho_n^{(t)} M + \sum_{k \neq n} \Pi_{kn} \rho_k^{(t)} M - \sum_{k \neq n} \Pi_{nk} \rho_n^{(t)} M$$

- $\rho_n^{(t+1)} M$: # OF SYSTEMS IN STATE n
- $\rho_n^{(t)} M$: # OF SYSTEMS IN STATE n
- $\sum_{k \neq n} \Pi_{kn} \rho_k^{(t)} M$: # SYSTEMS GOING TO STATE n
- $\sum_{k \neq n} \Pi_{nk} \rho_n^{(t)} M$: # SYSTEMS LEAVING STATE n (NOTE THE MINUS)

BUT AT EQUILIBRIUM $\rho_n^{(t+1)} = \rho_n^{(t)} = \rho_n$

HENCE

$$\cancel{\rho_n^{(t+1)}} = \cancel{\rho_n^{(t)}} + \sum_{\substack{k \\ k \neq n}} \Pi_{kn} \rho_k^{(t)} \rho_k - \sum_{\substack{k \\ k \neq n}} \Pi_{nk} \rho_n^{(t)} \rho_k$$

THUS HAVING

$$\sum_{\substack{k \\ k \neq n}} \Pi_{kn} \rho_k - \sum_{\substack{k \\ k \neq n}} \Pi_{nk} \rho_k = 0$$

$$\sum_{\substack{k \\ k \neq n}} \Pi_{kn} \rho_k + \Pi_{mn} \rho_n = \sum_{\substack{k \\ k \neq n}} \Pi_{nk} \rho_n + \Pi_{nn} \rho_n$$

$$\sum_k \Pi_{kn} \rho_k = \sum_k \Pi_{nk} \rho_n$$

BUT $\sum_k \Pi_{nk} = 1$ HENCE

$$\sum_k \rho_k \Pi_{kn} = \rho_n$$

WHICH CAN BE WRITTEN IN MATRIX FORM AS FOLLOWS

$$\vec{\rho}\, \Pi = \vec{\rho} \quad (1)$$

WHERE $\vec{\rho} = (\rho_1 \ldots \rho_k \ldots)$

AND Π IS A MATRIX WHOSE ELEMENTS ARE AND $(\Pi)_{nk} = \Pi_{nk}$

FOR AN ERGODIC MARKOV CHAIN WITH A FINITE SPACE OF STATES IF WE CONSIDER $\vec{\rho}\, \Pi = \lambda \vec{\rho}$ $\exists$ UNIQUE ρ SUCH THAT $\lambda = 1$, Π IS NOT GUARANTEED TO BE DIAGONALIZABLE AND THE OTHER EIGENVALUES λ_k ARE SUCH $|\lambda_k| < 1$ (THEY CAN BE OBTAINED BY JORDAN DECOMP.)

EQ. (1) IS CALLED BALANCE CONDITION AND IT STATES THAT VECTOR $\vec{p}$ IS AN EIGEN VECTOR OF MATRIX Π WITH EIGENVALUES 1.

EQ. (1) TELLS US THAT THE NUMBER OF SYSTEMS UNDERGOING A TRANSITION TO STATE n IS EQUAL TO THE NUMBER OF SYSTEMS LEAVING STATE n

WE CAN IMPOSE A STRONGER CONDITION

SINCE

$$\sum_k \Pi_{kn} p_k = \sum_k \Pi_{nk} p_n$$

THAT IS

$$\sum_k \left(\Pi_{kn} p_k - \Pi_{nk} p_n \right) = 0$$

WHICH IS SATISFIED IF WE REQUIRE

$$\Pi_{kn} p_k = \Pi_{nk} p_n \quad (2)$$

WHICH IS CALLED "DETAILED BALANCE CONDITION"

IN THIS CASE Π IS DIAGONALIZABLE WITH REAL EIGENVALUES

THIS CONDITION STATES THAT THE NUMBER OF SYSTEMS GOING FROM STATE k TO STATE n HAS TO BE EQUAL TO THE NUMBER OF SYSTEMS GOING FROM STATE n TO STATE k.

IN FOLLOWING $k \to 0$ SO THAT EQ. (2) BECOMES

$$\rho_0 \Pi_{0n} = \rho_n \Pi_{n0} \tag{3}$$

SO THAT WE CAN THINK THAT 0 STANDS FOR "OLD" AND n FOR "NEW".

ACCORDING TO THIS CONDITION WE CAN IMPOSE THE EQUILIBRIUM DISTRIBUTION REACHED BY THE SYSTEM BY PROVIDING A TRANSITION PROBABILITY WHICH ENFORCES EQ. (3) GIVEN A CERTAIN PROBABILITY DISTRIBUTION ρ_k

LET'S FIND OUT A WAY TO PROVIDE Π_{on} (MONTE CARLO METROPOLIS)

$$\Pi_{on} = \alpha(o \to n)\, acc(o \to n)$$

$\alpha(o \to n)$: TRIAL MOVE

$acc(o \to n)$: ACCEPTANCE PROBABILITY

$\alpha(o \to n)$ UPDATE SYSTEM'S STATE (I.E. PARTICLES POSITIONS) RANDOMLY

AND WE REQUIRE

I) $\alpha(o \to n) = \alpha(n \to o)$

I.E. THAT IT IS SYMMETRIC (EVEN IF IT IS NOT NECESSARY, AS WE SHALL SEE WHEN WE WILL DISCUSS BIASED MONTE CARLO)

II) $$acc(o \to n) = \min\left\{1, \frac{\rho_n}{\rho_o}\right\}$$

NOTE: IF $\rho = \frac{e^{-\beta V}}{Z}$ THEN

$$\frac{\rho_n}{\rho_o} = e^{-\beta(V_n - V_o)} = e^{-\beta \Delta V}$$

WHERE ΔV IS THE VARIATION OF POTENTIAL ENERGY AFTER THE TRIAL MOVE.

BY ENFORCING I) AND II)
DETAILED BALANCE CONDITION
IS FULLFILLED, INDEED

CASE I

CONSIDER A MOVE SUCH THAT

$\rho_n < \rho_o$ THEN

$$\rho_o \Pi_{on} = \rho_n \Pi_{no}$$

$$\rho_n \cancel{\alpha(o \to n)}\, acc(o \to n) = \rho_n \cancel{\alpha(n \to o)}\, acc(n \to o)$$

$$\rho_o \underbrace{\min\left(1, \frac{\rho_n}{\rho_o}\right)}_{<1} = \rho_n \underbrace{\min\left(1, \frac{\rho_o}{\rho_n}\right)}_{>1}$$

$$= \frac{\rho_n}{\rho_o} \qquad = 1$$

$$\cancel{\rho_o}\ \cancel{\frac{\rho_n}{\rho_o}} = \rho_n \Rightarrow \rho_n = \rho_n$$

SIMILARLY ONE CAN PROCEED
BY ASSUMING $\frac{\rho_n}{\rho_o} > 1$

CASE II

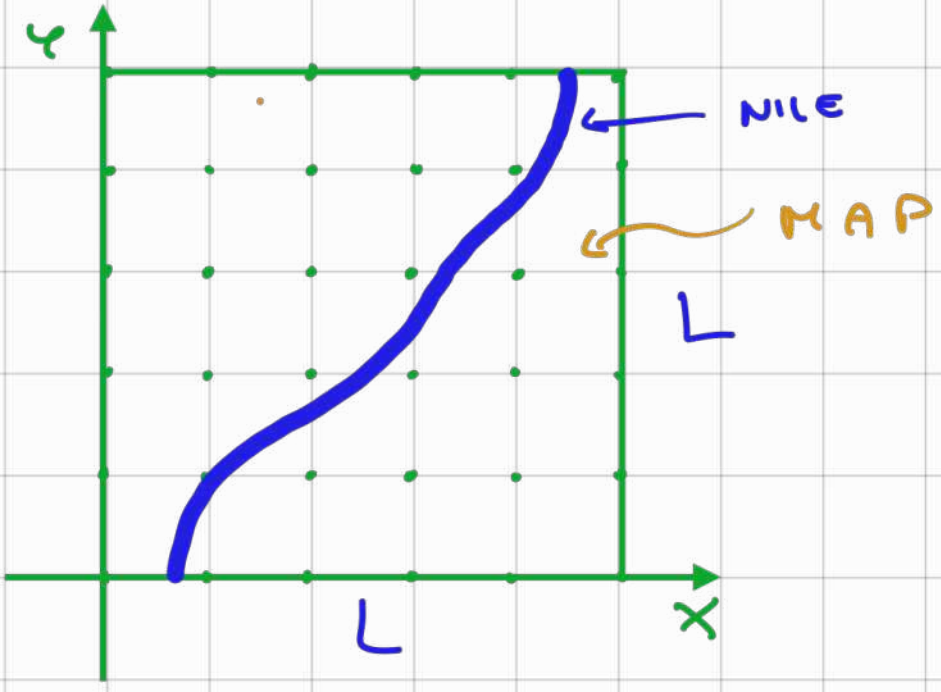
$\rho_n > \rho_o$

$$\rho_0 \underbrace{\min\left(1, \frac{\rho_n}{\rho_0}\right)}_{>1} = \rho_n \underbrace{\min\left(1, \frac{\rho_0}{\rho_n}\right)}_{<1}$$

$$= 1 \qquad = \rho_0/\rho_n$$

$$\rho_0 = \rho_n \frac{\rho_0}{\rho_n} \Rightarrow \rho_0 = \rho_0$$

A NEAT EXAMPLE:
CALCULATION OF AVERAGE DEPTH OF NILE



SOME DEFINITIONS

$A = L^2 = \{$ MAP AREA $\}$

$$\Theta(x,y) = \begin{cases} 1 & \text{IF } (x,y) \in \text{NILE} \\ 0 & \text{OTHERWISE} \end{cases}$$

$d(x,y) = \{$ DEPTH OF TERRAIN AT $(x,y)\}$

NOTE THAT IT CAN BE $\neq 0$ ALSO OUTSIDE NILE

HENCE

$$\mathcal{S} = \int_A \theta(x,y)\, dx\, dy = \{\text{AREA OF NILE}\}$$

AND

$$\langle d(x,y) \rangle = \int_A \frac{d(x,y)\,\theta(x,y)}{\mathcal{S}}\, dx\, dy$$

WE CAN CALCULATE THIS INTEGRAL THROUGH A STANDARD QUADRATURE PROCEDURE (E.G. RECTANGLES METHOD)

... OR WE CAN USE A METROPOLIS MC

$\frac{\theta(x,y)}{\mathcal{S}}$ IS A PROBABILITY DISTRIBUTION

INDEED $\int_A \frac{\theta(x,y)}{\mathcal{S}}\, dx\, dy = 1$

WHERE $\mathcal{S}$ ANALOGOUS OF PARTITION FUNCTION

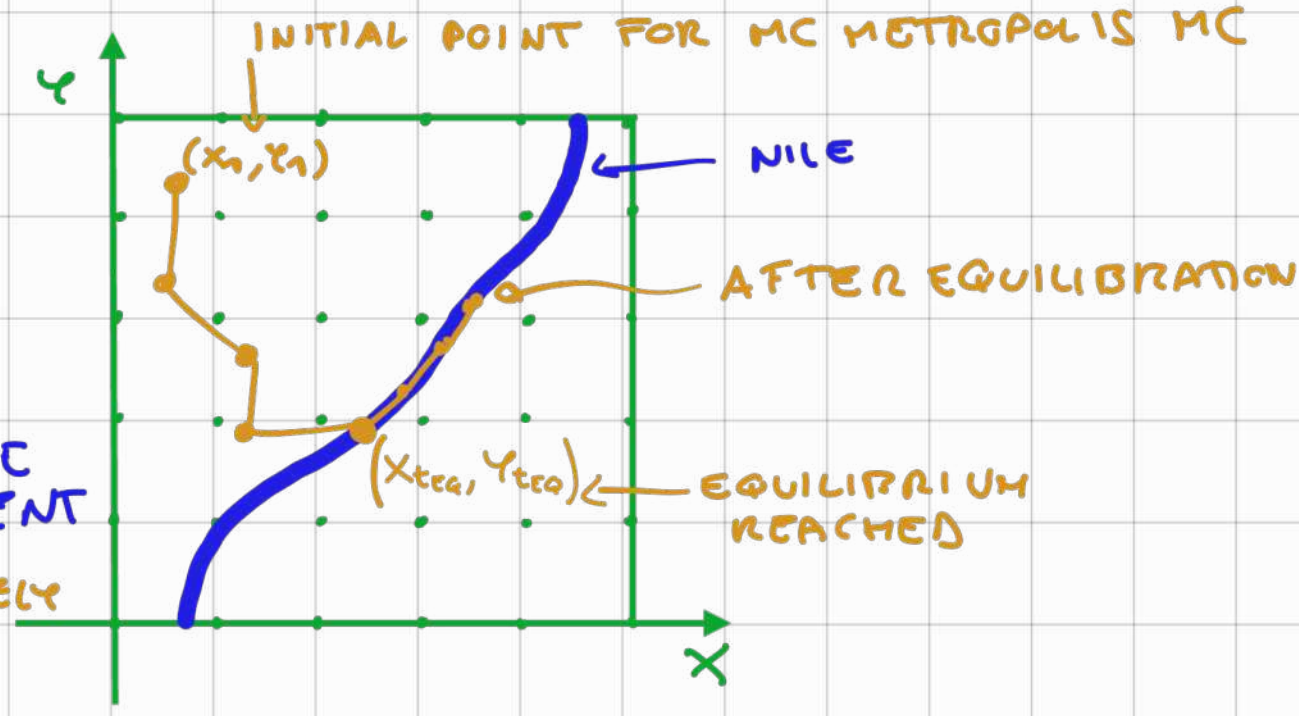
AND $d(x,y)$ IS AN OBSERVABLE SO THAT

$\int_A d(x,y)\, \frac{\theta(x,y)}{\mathcal{S}}\, dx\, dy$ CAN BE VIEWED AS A ENSEMBLE AVERAGE AND WE CAN EMPLOY A METROPOLIS MC

EACH TRIAL MOVE CONSISTS IN RANDOMLY UPDATING THE POSITION OF THE CURRENT POINT (I.E. LOCATION ON THE MAP) (x_0, y_0) TO A NEW POINT (x_n, y_n)

AND REJECTING THIS TRIAL MOVE IF $(x_o, y_o) \in$ NILE AND $(x_n, y_n) \notin$ NILE

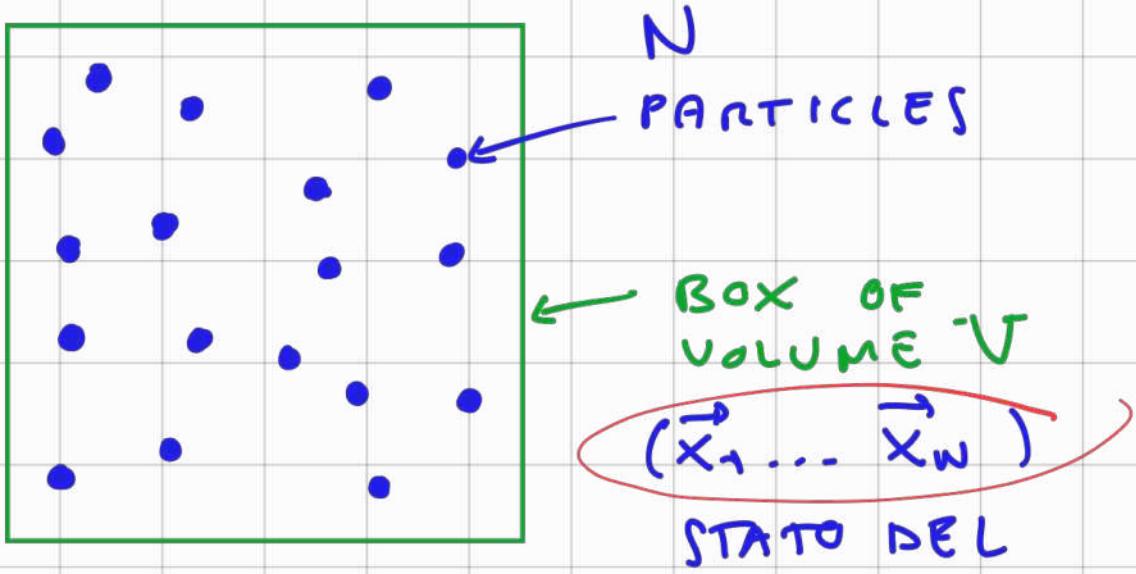
WE WILL END UP WITH A RANDOM WALK AS FOLLOWS :



NOTE THAT "HIT AND MISS" WOULD BE INEFFICIENT

VERY UNLIKELY TO HIT THE NILE

METROPOLIS MC FOR A SYSTEM OF N PARTICLES



I) TRIAL MOVE $\alpha(o \to n)$

CHOOSE RANDOMLY A PARTICLE AND DISPLACE IT BY

$$\Delta\vec{r} = (\Delta x, \Delta y, \Delta z)$$

1 MC STEP ≡ N TRIAL MOVE

WHERE

$$\Delta x = \xi_x \delta$$
$$\Delta y = \xi_y \delta$$
$$\Delta z = \xi_z \delta$$

ξ_α ARE RANDOM VARIABLES SUCH THAT

$$-1 < \xi_\alpha < 1$$

WITH $\alpha \in \{x, y, z\}$

II) ACCEPTANCE $\alpha(o \to n)$

$$acc(o \to n) = \min\left\{1, \frac{e^{-\beta U_n}}{e^{-\beta U_o}}\right\} =$$

$$= \min\left\{1, e^{-\beta \Delta U}\right\}$$

HENCE IF $\Delta U < 0 \Rightarrow$ ACCEPTED

OTHERWISE

ξ RANDOM VARIABLE $0 \leq \xi < 1$

IF $\xi \leq e^{-\beta \Delta U}$ ACCEPTED

$\xi > e^{-\beta \Delta U}$ REJECTED

EFFICIENCY

$$\eta = \{\text{EFFICIENCY}\} \equiv \frac{\sum_{k=1}^{L} \delta r_k^2}{\tau} = \frac{\Delta r^2}{\tau}$$

SUM OVER SINGLE PARTICLE MOVES ($\delta r_k = 0$ IF REJECTED)

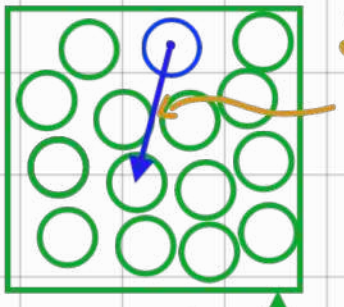
Δr^2 QUANTIFIES THE "TOTAL DISPLACEMENT", I.E. HOW MUCH THE INITIAL CONFIGURATION CHANGED

$\tau = \{$ COMPUTING TIME FOR CARRYING OUT L TRIAL MOVES $\} = t_1 L$

t_1: TIME NEEDED TO MOVE ONE PARTICLE (I.E. TO CALCULATE ΔU)

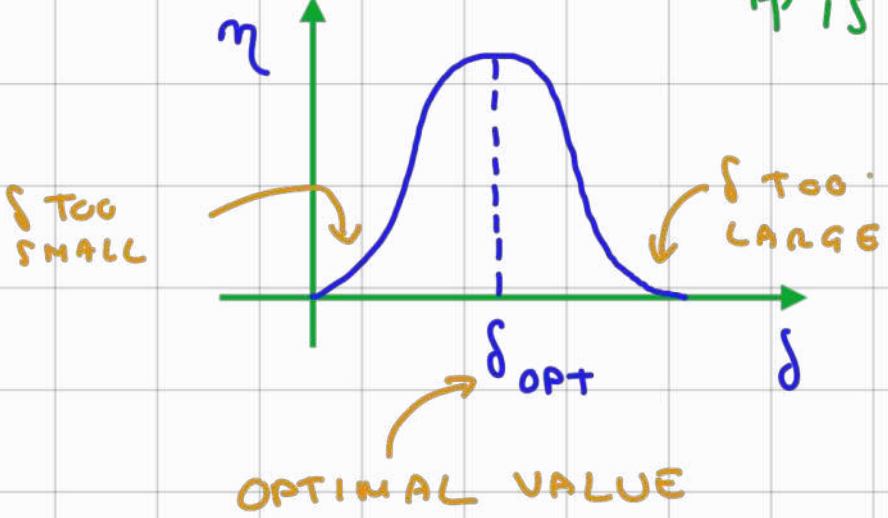
I) IF $\delta \to 0$ $\Delta r^2 \to 0$ BUT $\tau = t_1 \cdot L$ DOES NOT $\Rightarrow \eta \to 0$

II) IF δ IS LARGE (E.G. LARGER THAN PARTICLE SIZE)



δ LARGE, IT IS VERY LIKELY TO HAVE AN OVERLAP SO THAT ENERGY IS LARGE AND MOVE IS REJECTED

HENCE $\Delta r^2 \to 0$ BUT AGAIN τ IS FINITE $\Rightarrow \eta \to 0$



BE $R = \{\text{ACCEPTANCE RATE OF TRIAL MOVES}\} =$

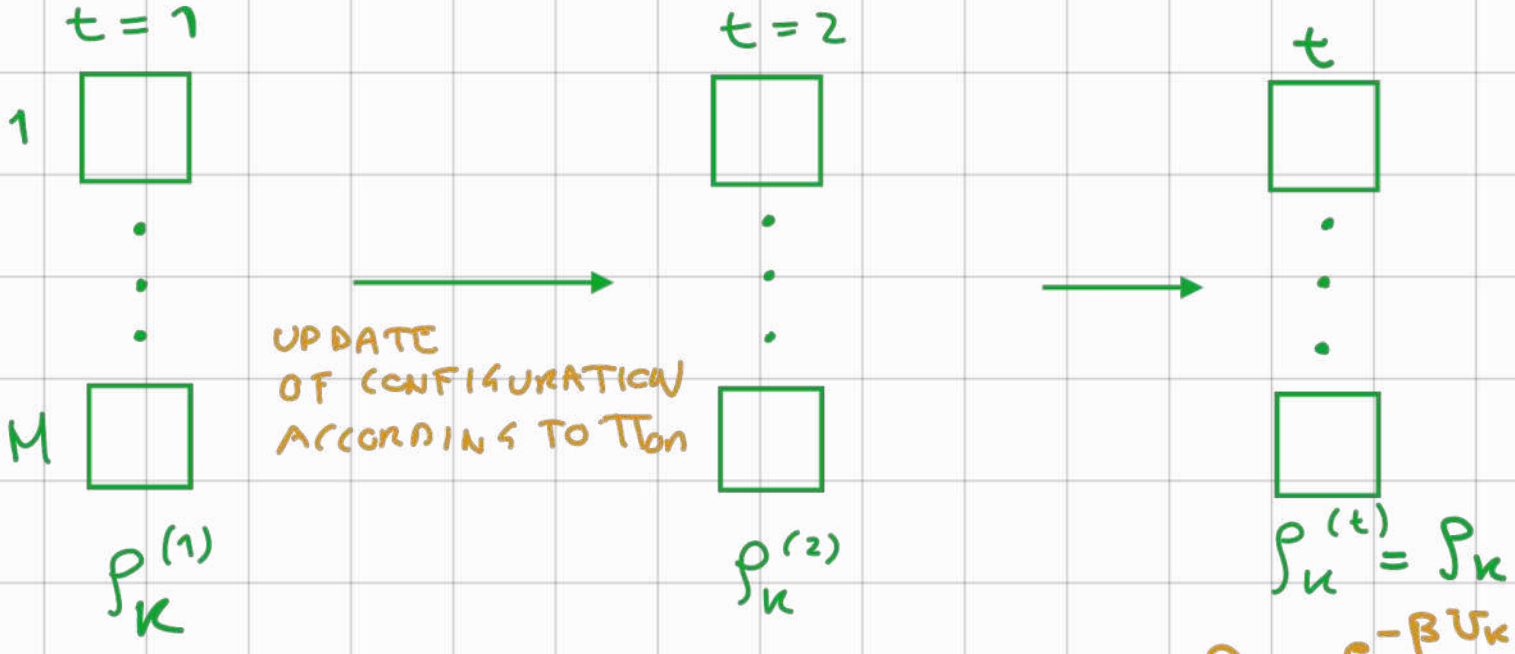
$$= \frac{\{\#\text{ ACCEPTED TRIAL MOVES}\}}{\{\#\text{ TRIAL MOVES}\}}$$

IF $0.3 < R < 0.5 \Rightarrow \delta \approx \delta_{OPT}$

(EMPIRICAL RULE)

Q. WHAT R DO WE HAVE IF WE MOVE N PARTICLES AT A TIME?

CONSIDER AGAIN M IDENTICAL SYSTEMS BEING SIMULATED IN THE NTV ENSEMBLE (CANONICAL ENSEMBLE)



$$\rho_K = \frac{e^{-\beta U_K}}{Z}$$

EQUILIBRIUM

AFTER A TRIAL MOVE OF 1 PARTICLE

$$\langle \Delta U \rangle = \left\langle \sum_\alpha \frac{\partial U}{\partial r_i^\alpha} \delta r_i^\alpha + \right. \qquad \alpha = x, y, z$$

$$\left. + \frac{1}{2} \sum_{\alpha,\beta} \frac{\partial^2 U}{\partial r_i^\alpha \partial r_i^\beta} \delta r_i^\alpha \delta r_i^\beta \right\rangle =$$

WHERE $\langle \dots \rangle$ IS OVER ALL M SYSTEMS

$$= \left\langle \sum_\alpha \frac{\partial U}{\partial \eta_i^\alpha} \right\rangle \overline{\delta\eta_i^\alpha} + \left\langle \frac{1}{2} \sum_{\alpha\beta} \frac{\partial^2 U}{\partial \eta_i^\alpha \partial \eta_i^\beta} \right\rangle \overline{\delta\eta_i^\alpha \, \delta\eta_i^\beta}$$

(overline: AVERAGE OVER TRIAL MOVES; $\langle\rangle$: AVERAGE OVER M RUNS)

↑ WE SPLITTED THE AVERAE SINCE AVERAGE OVER TRIAL MOVES ($\overline{x}$) IS INDIPENDENT OF AVERAGE OVER M SYSTEMS ($\langle x \rangle$)

BUT $\overline{\delta\eta_i^\alpha} = 0$ AND

$\overline{\delta\eta_i^\alpha \, \delta\eta_i^\beta} = 0$ IF $\alpha \neq \beta$

AND $\overline{(\delta\eta_i^x)^2} = \overline{(\delta\eta_i^y)^2} = \overline{(\delta\eta_i^z)^2} = \overline{\delta\eta^2}$

MOREOVER $\frac{1}{2} \sum_\alpha \frac{\partial U}{\partial \eta_i^\alpha \partial \eta_i^\alpha} \equiv f(U)$

HENCE WE ARE LEFT WITH

$$= f(U) \, \overline{\delta\eta^2}$$

A MOVE IS LIKELY ACCEPTED IF

$\langle \Delta U \rangle \simeq k_B T$ THUS HAVING

$$f(U) \, \overline{\delta\eta^2} = k_B T \Rightarrow \overline{\delta\eta^2} = \frac{k_B T}{f(U)}$$

THAT IS IF WE MOVE 1 PARTICLE AT A TIME L TIMES:

$$r_1 = \frac{\cancel{L} \, \overline{\delta\eta^2}}{\cancel{L} \, t_1} = \frac{\overline{\delta\eta^2}}{t_1} = \frac{k_B T}{t_1 f(U)}$$

IF WE ATTEMPT TO MOVE N PARTICLES AT A TIME

$$\langle \Delta U \rangle = f(U)\, \overline{\delta r^2}\, N$$

SINCE WE HAVE CONTRIBUTIONS FROM N PARTICLES (NOT JUST i-TH PARTICLE AS BEFORE)

HENCE:

$$\overline{\delta r^2} = \frac{k_B T}{N f(U)}$$

→ AVERAGE DISPLACEMENT OF A SINGLE PARTICLE AFTER A TRIAL MOVE

NOTE THAT $1/N$ $\eta = \frac{\sum_k \delta r_k^2}{T}$

$\sum_k \delta r_k^2$ IS NOW A SUM OVER ALL PARTICLE DISPLACEMENTS AFTER L TRIAL MOVES, I.E.:

$$\sum_k \delta r_k^2 = L \cdot N \cdot \overline{\delta r^2}$$

HENCE THE EFFICIENCY η_N OF MOVING SIMOULTANEOUSLY N PARTICLES AFTER L TRIAL MOVES IS

$$\eta_N = \frac{\overline{\delta r^2}\, \cancel{K N}}{t_1 \cancel{K N}} = \frac{k_B T}{N t_1 f(U)}$$

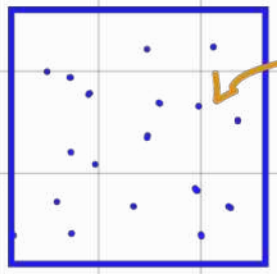
← η_1

SO THAT

$$\eta_N = \frac{\eta_1}{N}$$

I.E. η_N IS N TIMES SMALLER THAN η_1!

INTERACTION POTENTIALS



2 BODY INTERACTIONS

$$U = \frac{1}{2}\sum_{i\neq j} U_2(\vec{r}_i, \vec{r}_j) + \sum_{i<j<k} U_3(\vec{r}_i, \vec{r}_j, \vec{r}_k)$$

3 BODY INTERACTIONS

$$= \frac{1}{2}\sum_{i\neq j} U_2(\vec{r}_{ij}) \qquad \vec{r}_{ij} = \vec{r}_i - \vec{r}_j$$

NOTE THAT

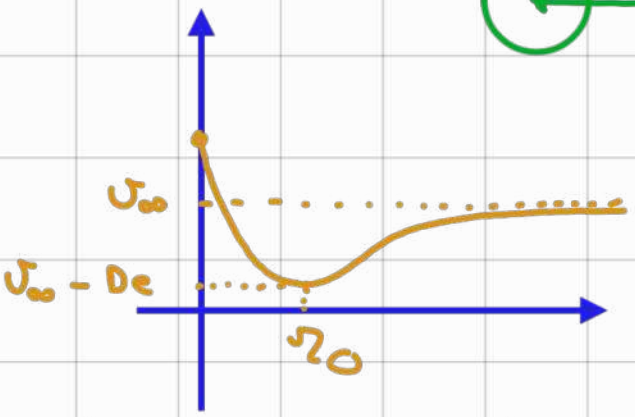
$$U = \frac{1}{2}\sum_{i\neq j} U_2(\vec{r}_{ij}) = \sum_{i<j} U_2(\vec{r}_{ij})$$

MORSE POTENTIAL (BIATOMIC MOLECULE)

$$U_2(r) = U_M(r) = U_\infty + D_e\left\{\left[1 - e^{-\alpha(r-r_0)}\right]^2 - 1\right\}$$



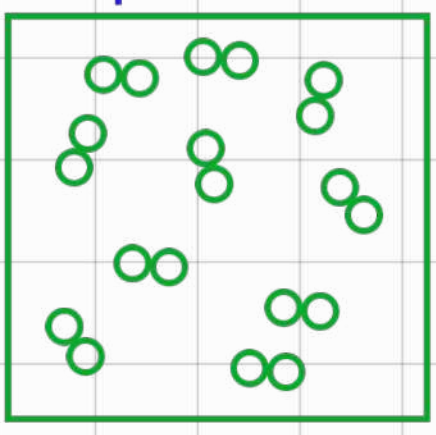
INTERACTION BETWEEN TWO ATOMS BELONGING TO A BIATOMIC MOLECULE



$$\left.\frac{dU_M}{dr}\right|_{r=r_0} = 0$$

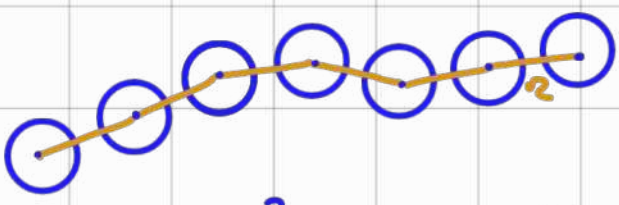
AND

$$U_M(r_0) = U_\infty - D_e$$



INTERACTION OF PAIR OF PARTICLES BELONGING TO A BIATOMIC MOLECULE

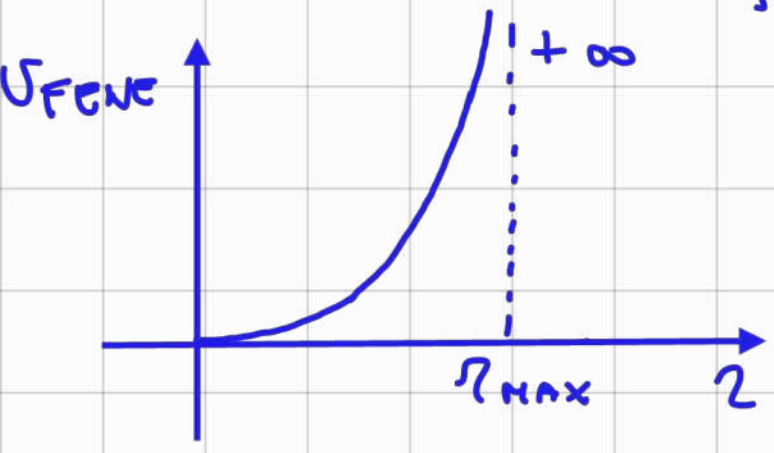
POLYMERS



$$U_{FENE}(r) = \begin{cases} -\frac{1}{2} r_{MAX}^2 \epsilon \ln\left[1 - \left(\frac{r}{r_{MAX}}\right)^2\right] & r \leq r_{MAX} \\ \infty & r > r_{MAX} \end{cases}$$

INTERACTIONS OF TWO SUCCESSIVE PARTICLES BELONGING TO A POLYMER

FENE = { FINITE EXTENSIBLE NON LINEAR ELASTIC }



IF $r \ll r_{MAX}$

$$U_{FENE} \approx \epsilon \frac{r^2}{2}$$

HENCE ϵ IS THE ELASTIC CONSTANT OF SMALL OSCILLATIONS

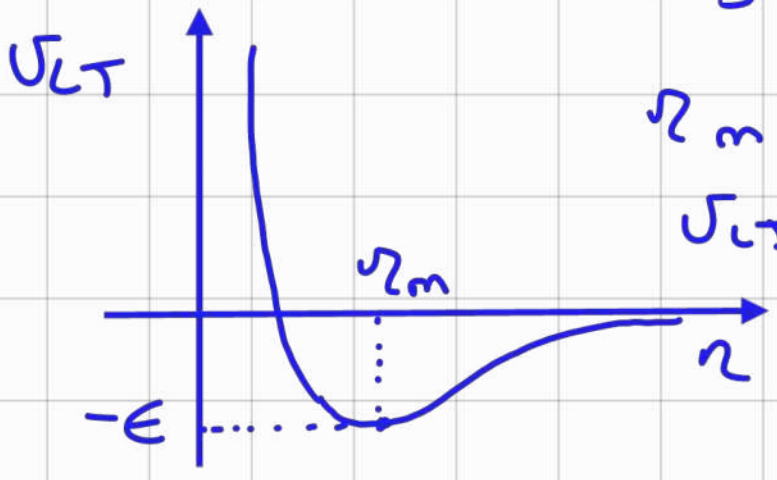
LENNARD-JONES

(NOBLE GASE SUCH AS ARGON, NEON, XEON OR RADON)

↑ MONOATOMIC GASES

[1964 RAMAN AND 1969 VERLET AND HANSEN]

$$U_{LJ}(r) = 4\epsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$



$$r_m = 2^{1/6}\sigma \approx 1.12\sigma$$

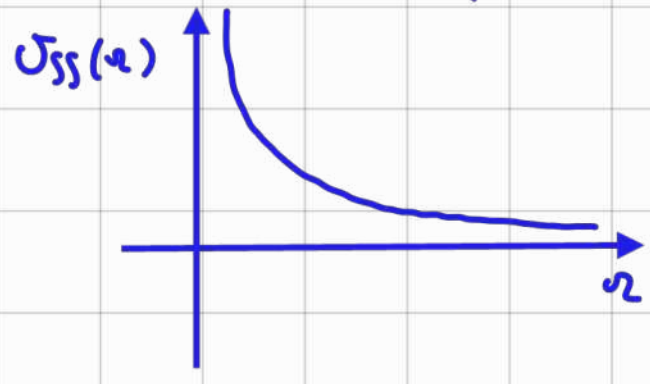
$$U_{LJ}(r_m) = -\epsilon$$

THIS POTENTIAL WELL DESCRIBES INTERACTION BETWEEN UNCHARGED ATOMS

SOFT SPHERE POTENTIAL

$$U_{SS}(r) = \varepsilon\left(\frac{\sigma}{r}\right)^{n}$$

n QUANTIFIES SOFTNESS OF PARTICLES



SCALING LAWS OF SOFT-SPHERE FLUID

$$F = -k_B T \ln Z \qquad Z = \int e^{-\beta U_{SS}} d^{3N}r$$

$$U_{SS} = \varepsilon \sum_{i<j} \left(\frac{\sigma}{r_{ij}}\right)^n \qquad r_{ij} = |\vec{r}_{ij}|$$

$$Z = \int e^{-\beta\varepsilon \sum_{i<j}\left(\frac{\sigma}{r_{ij}}\right)^n} d^{3N}r$$

$\vec{r}_i = V^{1/3}\vec{\xi}_i$ SCALED COORDINATES

$$= \int e^{-\frac{\beta\varepsilon\sigma^n}{V^{n/3}} \sum_{i<j}\left(\frac{1}{\xi_{ij}}\right)^n} V^N d^{3N}\xi$$

$\frac{N}{V} = \rho$

CONSIDER $\dfrac{\beta\varepsilon\sigma^n}{V^{n/3}} = \left[(\beta\varepsilon)^{3/n}\sigma^3 \dfrac{N}{V}\right]^{n/3} \dfrac{1}{N^{n/3}}$

IF WE DEFINE $\Gamma \equiv (\beta\varepsilon)^{3/n}\sigma^3\rho = \Gamma(T,\rho)$

WE HAVE

$$Z = V^N \int e^{-\left(\frac{\Gamma}{N}\right)^{n/3} \sum_{i<j}\left(\frac{1}{\xi_{ij}}\right)^n} d^{3N}\xi$$

$$= Z_{ID} \cdot Z_{EX}(\Gamma)$$

WHERE $Z_{ID} = V^N$ AND

$$Z_{EX}(\Gamma) = \int e^{-(\Gamma/N)^{n/3} \sum_{i<j}\left(\frac{1}{\xi_{ij}}\right)^n} d^{3N}\xi$$

$U = U_{ID} + U_{EX} \qquad U_{ID} = \langle K \rangle = \frac{3}{2} N k_B T$

$$\beta U_{EX} = \beta \langle U_{SS} \rangle = \beta \int U_{SS} \frac{e^{-\beta U_{SS}}}{Z} d^{3N}r =$$

$$= \int \beta U_{SS} \frac{e^{-\beta U_{SS}}}{V^N Z_{EX}(\Gamma)} V^N d^{3N}\xi$$

$$\overline{\beta U_{ss}} = \left(\frac{\Gamma}{N}\right)^{n/3} \sum_{i<j} \left(\frac{1}{r_{ij}}\right)^n = f(\Gamma)$$

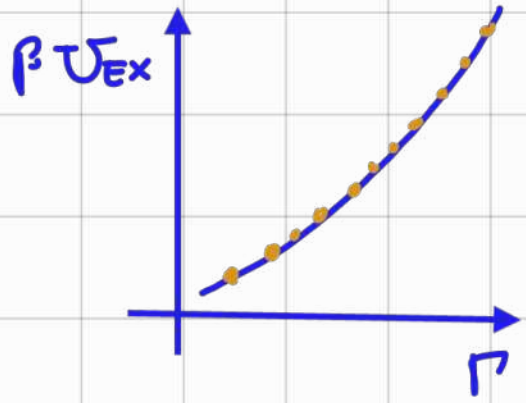
↳ FOR FIXED N

HENCE

$$\beta U_{ex} = F(\Gamma)$$

SIMULATIONS WITH SAME $\Gamma = \Gamma(T, \rho)$
(EVEN IF T AND ρ ARE DIFFERENT)

WILL YIELD SAME VALUE OF βU_{ex}, I.E.



• SIMULATIONS

SCALING LAW

NOTE THAT:
$F = -k_B T \ln Q$ BUT
$Q = \frac{\Lambda^{-3N}}{N!} Z$ AND $\frac{\Lambda^{-3N}}{N!}$ DOES NOT DEPEND ON V

SIMILAR SCALING FOR PRESSURE

$$P = -\frac{\partial F}{\partial V} = \frac{\partial (k_B T \ln Z)}{\partial V}$$

BUT $Z = Z_{ID} \cdot Z_{ex}$, $Z_{ID} = V^N$

$$= \underbrace{\frac{\partial}{\partial V}\left[k_B T N \ln V\right]}_{P_{ID}} + \underbrace{k_B T \frac{1}{Z_{ex}} \frac{\partial Z_{ex}}{\partial V}}_{P_{ex}}$$

$$P = P_{ID} + P_{ex} = \frac{\rho}{\beta} + P_{ex}$$

$$P_{ID} = \rho k_B T$$

$$P_{ex} = k_B T \frac{1}{Z_{ex}} \frac{\partial Z_{ex}}{\partial V} = k_B T \frac{1}{Z_{ex}} \frac{\partial Z_{ex}}{\partial \Gamma} \frac{\partial \Gamma}{\partial V}$$

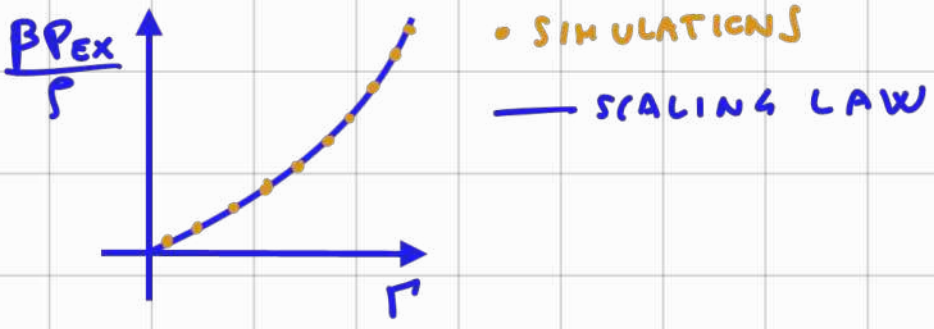
BUT $\frac{\partial \Gamma}{\partial V} = -(\beta\varepsilon)^{3/n} \frac{N}{V^2} \sigma^3 = \Gamma \frac{\rho}{N}$

$$= \rho k_B T \underbrace{\frac{\Gamma/N}{Z_{ex}(\Gamma)} \frac{\partial Z_{ex}}{\partial \Gamma}}_{G(\Gamma)} \leftarrow \text{FOR FIXED } N$$

$$\frac{\beta P_{ex}}{\rho} = G(\Gamma) \quad \text{SCALING LAW}$$

IF ONE PERFORMS SIMULATIONS CHANGING T AND ρ

BUT AT FIXED Γ ONE HAS TO GET



COULOMB INTERACTIONS

$$U_C(r) = \frac{1}{4\pi \varepsilon_r \varepsilon_0} \frac{q_1 q_+}{r}$$

SCREENED INTERACTIONS [DEBYE-HUCKEL THEORY]

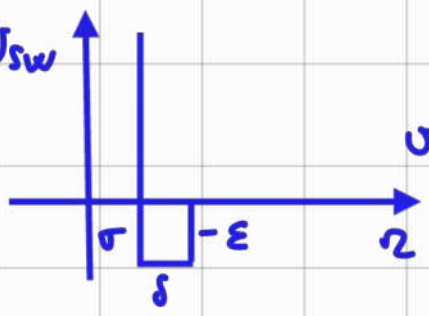
$$U_{SC}(r) = A \frac{q_1 q_+}{r} e^{-r/\lambda}$$

λ IS THE DEBYE SCREENING LENGTH

λ DEPENDS ON CHARGE IN SOLUTION

HARD SPHERES (ALREADY DISCUSSED)

SQUARE WELL



$$U_{SW} = \begin{cases} +\infty & r < \sigma \\ -\varepsilon & \sigma < r < \sigma + \delta \\ 0 & r > \sigma + \delta \end{cases}$$

TRUNCATION, SHIFT AND SMOOTHING OF INTERACTION POTENTIALS

INTERACTION POTENTIALS MAY EXTEND TO INFINITE DISTANCE (E.G. LENNARD-JONES), HENCE NEED OF TRUNCATION

$$U_{LJ}^{c}(r) = \begin{cases} 4\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right] & r \leq r_c \\ 0 & r > r_c \end{cases}$$

WHERE r_c IS THE CUTOFF RADIUS (IN OLD LITERATURE $r_c = 2.5\sigma$)

AT $r = r_c$ WE HAVE A DISCONTINUITY (BAD FOR MD SIMULATIONS AS WE SHALL SEE)

HENCE SHIFT OF POTENTIAL, I.E.

$$U_{LJ}^{cs}(r) = \begin{cases} U_{LJ}(r) - U_{LJ}(r_c) & r \leq r_c \\ 0 & r > r_c \end{cases}$$

NOTE THAT FOR MC SIMULATIONS SHIFT IS NOT NEEDED

MODIFIED LJ

TO HAVE ALSO SECOND DERIVATIVE CONTINUOUS

$$U_{LJ}^{M} = \begin{cases} U_{LJ}(r) + a \cdot r + b & r \leq r_c \\ 0 & r > r_c \end{cases}$$

WHERE a AND b ARE SUCH THAT

$$U_{LJ}(r_c) + a \cdot r_c + b = 0$$

$$\left.\frac{dU_{LJ}^{M}}{dr}\right|_{r=r_c} = \left.\frac{dU_{LJ}(r)}{dr}\right|_{r=r_c} + a = 0$$

SO THAT

$$a = -\left.\frac{dU_{LJ}}{dr}\right|_{r=r_c}$$

$$b = -U_{LJ}(r_c) - a\, r_c$$

TYPICALLY

$$2.5\,\sigma \leq r_c \leq 5\sigma$$

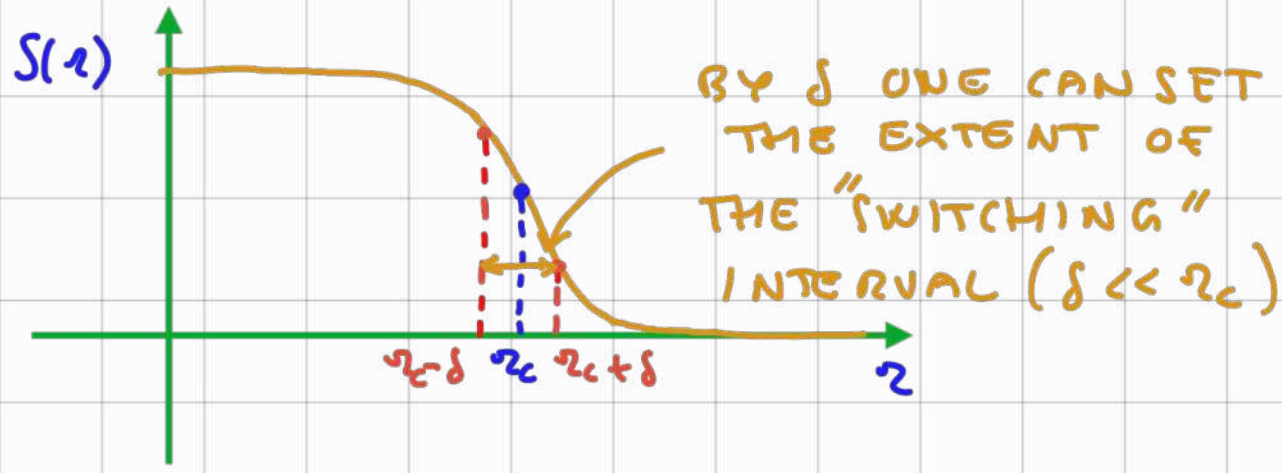
SWITCH FUNCTION

$$U_{LJ}^{S}(r) = U_{LJ}(r) \cdot S(r)$$

WHERE

$$S(r) = 0.5\left[1 + \tanh\left(\frac{r_c - x}{\delta}\right)\right]$$

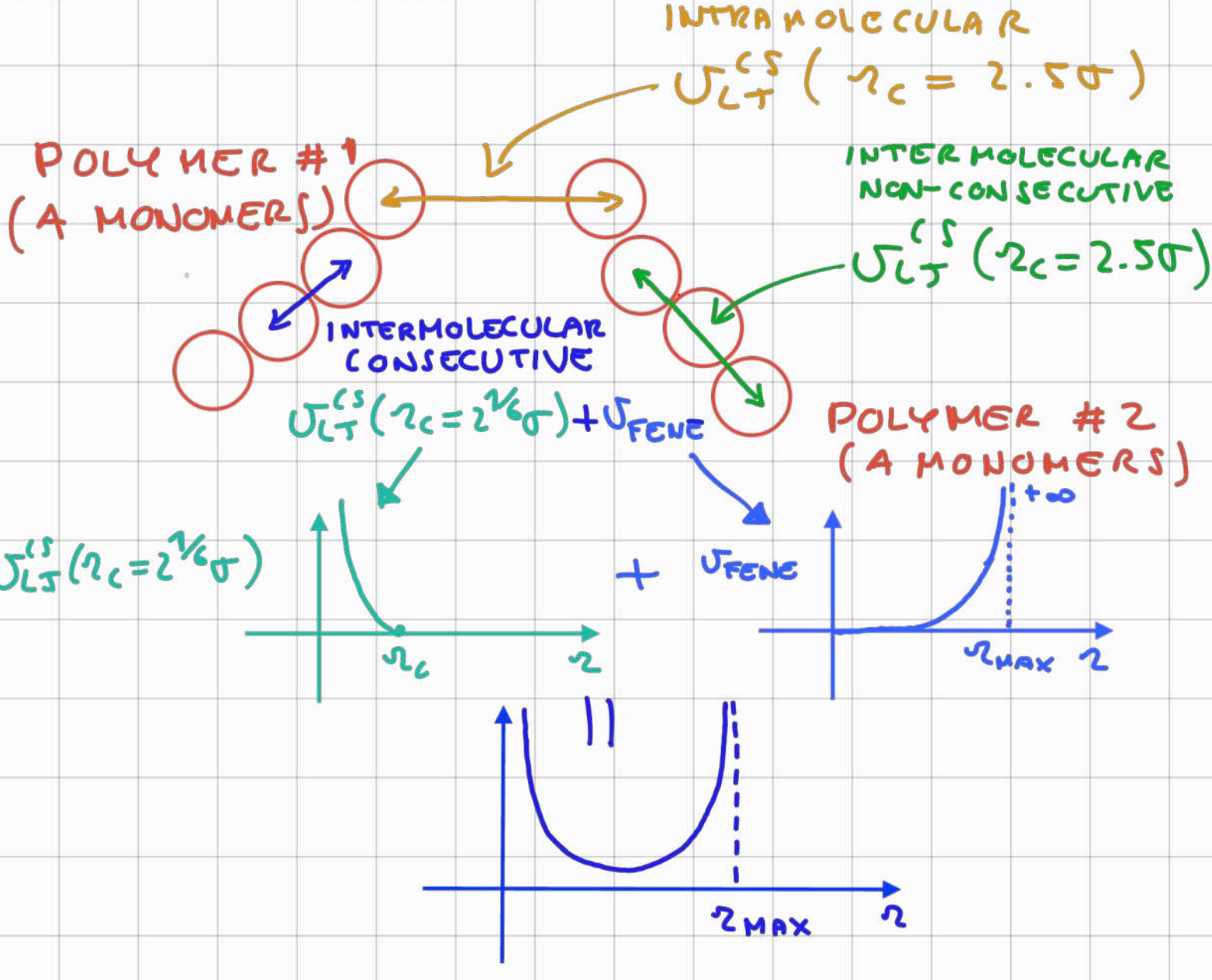
$$\tanh(x) = \frac{e^{x} - e^{-x}}{e^{x} + e^{-x}} \qquad \text{WITH} \quad \delta \ll r_c$$



LENNARD-JONES POTENTIAL FROM FIRST PRINCIPLES

(SEE SLIDES)

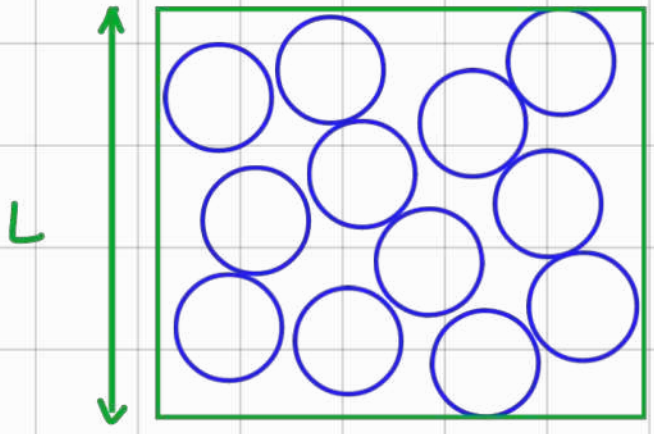
EXAMPLE SIMULATION OF A POLYMER MELT (KREMER-GREST MODEL)



MC ON LATTICE (E.G. ISING MODEL)

EACH MC CONSISTS IN UPDATING STATE OF A LATTICE SITE

SURFACE EFFECTS



$V = L^3$

N PARTICLES

$\rho = \frac{N}{V}$

$F = F_{SURF} + F_{BULK}$

$F_{SURF} = \gamma \sigma = \gamma \cdot 6 L^2$ (σ: AREA)

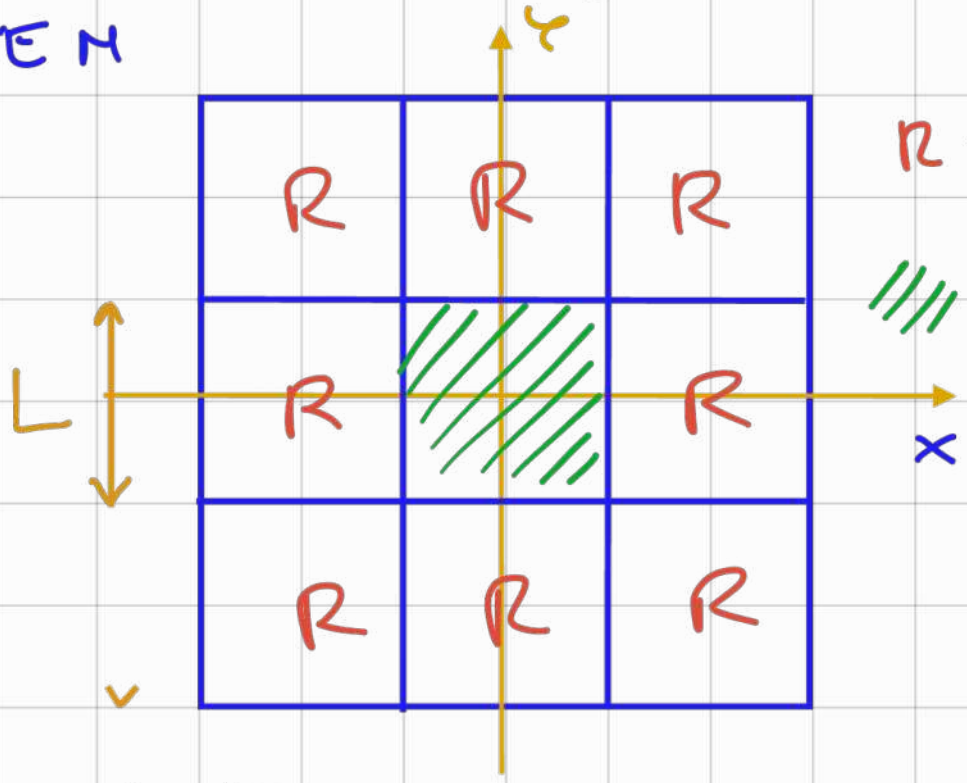
$F_{BULK} = k \cdot N = k \rho V = k \rho L^3$ (k: CONSTANT)

HENCE $\frac{F_{SURF}}{F_{BULK}} \sim \frac{1}{L}$

I.E. F_{SURF} IS NOT NEGLIGIBLE FOR SIMULATED SYSTEMS, SINCE BOX SIZE IS RATHER SMALL

NEED TO GET RID OF SURFACE EFFECTS TO COMPARE WITH EXPERIMENTAL RESULTS

INFINITE REPLICAS OF THE SYSTEM



R = REPLICAS

/// = ORIGINAL SYSTEM

$$U = \frac{1}{2} \sum_{i,j} U_2(\vec{r}_{ij})$$

↑ POTENTIAL ENERGY

$$\Downarrow$$

PAIR POTENTIAL ↓

$$U = \frac{1}{2} \sum_{\substack{i,j,\vec{n} \\ i \neq j,\ \vec{n}=0}} U_2\left(\vec{r}_i - (\vec{r}_j + L\vec{n})\right)$$

WHERE $\vec{n} = (n_x, n_y, n_z)$

n_α SIGNED INTEGER $\alpha = x, y, z$

PROBLEM: INFINITE TERMS IN CALCULATION OF ENERGY

$$\Downarrow$$

MINIMUM IMAGE CONVENTION

WHEN WE MOVE A PARTICLE i IN A MC SIMULATION WE NEED TO CALCULATE THE VARIATION OF U, I.E.

$$\Delta U_i = U_i' - U_i \qquad \vec{r}_i \to \vec{r}_i'$$

($\vec{r}_i'$: NEW STATE)

WHERE

$$U_i' = \sum_J v_2(\vec{r}_i' - \vec{r}_J)$$

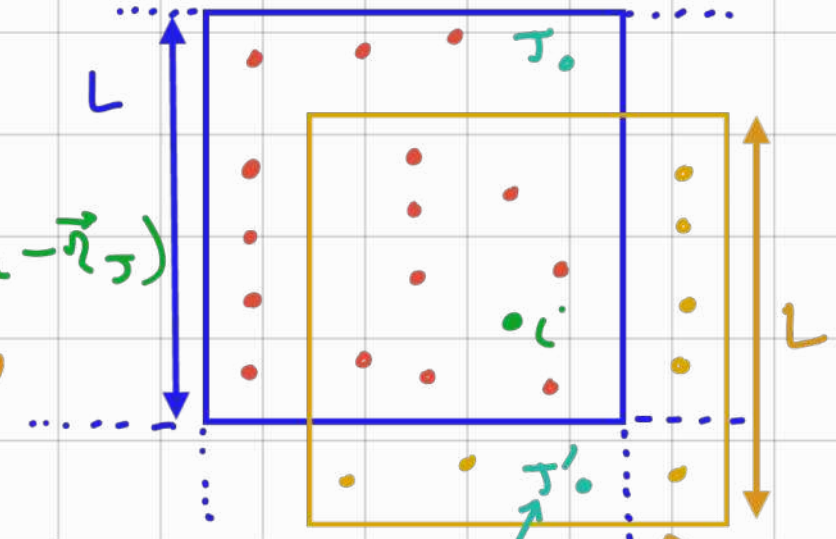
$$U_i = \sum_J v_2(\vec{r}_i - \vec{r}_J)$$

TO CALCULATE U_i AND U_i' WE EMPLOY THE MINIMUM IMAGE CONVENTION

ENERGY

$$U_i = \frac{1}{2}\sum_J v_2(\vec{r}_i - \vec{r}_J)$$

($\sum_J$: ALL J WITHIN BOX OF SIZE L CENTERED AT POSITION OF PARTICLE i)



WHERE WE REQUIRE $r_c < \frac{L}{2}$ SO THAT WE ARE NOT MISSING CONTRIBUTIONS

$$U_i = \sum_J v_2(\vec{r}_1 - \vec{r}_J) = \sum_J v_2(\vec{r}_{1J})$$

IF J IS NOT INSIDE MINIMUM IMAGE WE NEED TO CONSIDE IS PERIODIC IMAGE, I.E.

THIS AMOUNTS TO DO THIS REPLACEMENT

ELEMENT-WISE DIVISION

$$U_2(\vec{r}_{1J}) \rightarrow U_2\left(\vec{r}_{1J} - L \cdot \mathrm{RINT}\left(\frac{\vec{r}_{1J}}{L}\right)\right) \quad (*)$$

WHERE IF $\vec{U} = (U_x, U_y, U_z)$

$$\mathrm{RINT}(\vec{U}) \equiv \mathrm{RINT}\begin{pmatrix} U_x \\ U_y \\ U_z \end{pmatrix} \equiv \begin{pmatrix} \mathrm{RINT}(U_x) \\ \mathrm{RINT}(U_y) \\ \mathrm{RINT}(U_z) \end{pmatrix}$$

WHERE $\mathrm{RINT}(x) = \{$ INTEGER NEAREST TO $x\}$

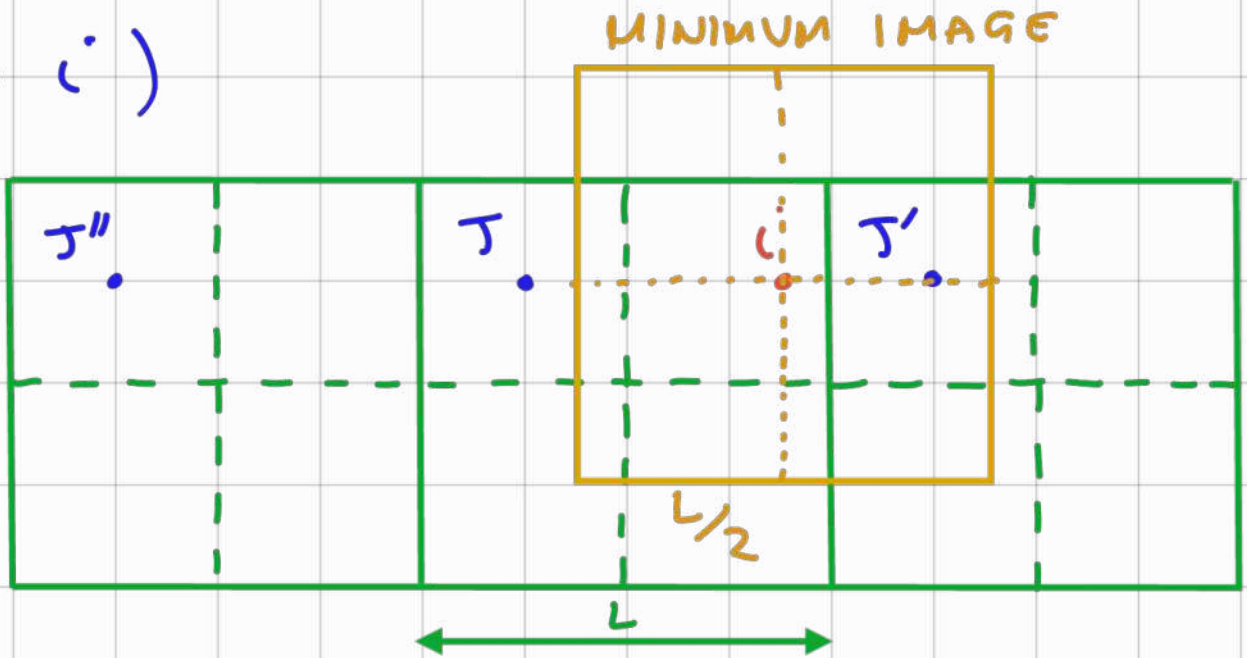
EXAMPLES

$\mathrm{RINT}(0.6) = 1$

$\mathrm{RINT}(0.2) = 0$

$\mathrm{RINT}(-0.7) = -1$

LET'S SHOW THAT (*) PROVIDES THE CORRECT MINIMUM IMAGE PARTICLE (I.E. THE PARTICLE J IN THE PERIODIC SYSTEM WHICH IS CLOSER TO i)



$$\vec{r}_i = (x_i, y_i) \qquad \vec{r}_j = (x_j, y_j)$$

$$\vec{r}_i - \vec{r}_j - L\,\mathrm{RINT}\left(\frac{\vec{r}_i - \vec{r}_j}{L}\right) =$$

BUT

$$L\,\mathrm{RINT}\left(\frac{\vec{r}_i - \vec{r}_j}{L}\right) = L\,\mathrm{RINT}\begin{pmatrix} \frac{x_i - x_j}{L} \\ \frac{y_i - y_j}{L} \end{pmatrix} =$$

ELEMENT WISE DIVISION

$$= L\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} L \\ 0 \end{pmatrix}$$

HENCE

$$\begin{pmatrix} x_i - x_j - L \\ y_i - y_j \end{pmatrix} = \begin{pmatrix} x_i - (x_j + L) \\ y_i - y_j \end{pmatrix} =$$

$$= \begin{pmatrix} x_i - x_{j'} \\ y_i - y_j \end{pmatrix} = \vec{r}_i - \vec{r}_{j'}$$

CLOSEST PERIODIC IMAGE OF J

FINAL REMARK:

ADJUSTEMENT OF DISTANCES BETWEEN PARTICLES (EQ. (*)) MUST BE APPLIED EVERY TIME ONE HAS TO CALCULATE DISTANCES FOR CALCULATION OF INTERACTION POTENTIAL

RADIAL DISTRIBUTION FUNCTION

$\tilde{\rho}(\vec{r}_1, \vec{r}_2)$ = { PROBABILITY OF FINDING A PARTICLE IN $\vec{r}_1$ AND ANOTHER ONE IN $\vec{r}_2$ }

EVENT A = { A PARTICLE IS IN $\vec{r}_1$ }

EVENT B = { // // // // $\vec{r}_2$ }

$$P(A \cap B) = \tilde{\rho}(\vec{r}_1, \vec{r}_2)$$

$$P(A \cap B) = P(A|B) \cdot P(B)$$

↑ CONDITIONAL PROBABILITY

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

NOTE THAT (HOMOGENEOUS CASE):
IF $P(A) = P(B)$
$P(A|B) = \frac{P(A \cap B)}{P(B)} =$
$= \frac{P(A \cap B)}{P(A)} = P(B|A)$

IF A AND B INDEPENDENT

$$P(A|B) = \frac{P(A)\,\cancel{P(B)}}{\cancel{P(B)}} = P(A)$$

GOING BACK TO $\tilde{\rho}$:

$$\tilde{\rho}(\vec{r}_1, \vec{r}_2) = \tilde{\rho}(\vec{r}_1|\vec{r}_2)\,\tilde{\rho}(\vec{r}_2) \qquad (*)$$

WE DEFINE:

$$g^{(2)}(\vec{r}_1, \vec{r}_2) \equiv \frac{\tilde{\rho}(\vec{r}_1, \vec{r}_2)}{\tilde{\rho}(\vec{r}_1)\,\tilde{\rho}_2(\vec{r}_2)}$$

} THIS IS 2-PARTICLE (PAIR) DISTRIBUTION FUNCTION

IF THE SYSTEM IS HOMOGENEOUS $\tilde{\rho}(\vec{r}_1) = \tilde{\rho}(\vec{r}_2) = \tilde{\rho}$

SO THAT:

$$g^{(2)}(\vec{r}_1, \vec{r}_2) = \frac{\tilde{\rho}(\vec{r}_1, \vec{r}_2)}{\tilde{\rho}^2}$$

BUT USING (*) WE HAVE:

$$g^{(2)}(\vec{r}_1, \vec{r}_2) = \frac{\tilde{\rho}(\vec{r}_1|\vec{r}_2)\,\cancel{\tilde{\rho}}}{\tilde{\rho}^{\cancel{2}}}$$

$$= \frac{\tilde{\rho}(\vec{r}_1 \mid \vec{r}_2)}{\tilde{\rho}}$$

SINCE SYSTEM IS HOMOGENEOUS WE CAN EXPECT

$g^{(2)}(\vec{r}_1, \vec{r}_2) = g^{(2)}(\vec{r}_1 - \vec{r}_2)$

SO THAT $g^{(2)}(\vec{r}_1 - \vec{r}_2) = g^{(2)}(\vec{r})$

AND $\tilde{\rho}(\vec{r}_1 \mid \vec{r}_2) = \tilde{\rho}(\vec{r})$

I.E. $\tilde{\rho}\, g^{(2)}(\vec{r}_1, \vec{r}_2) = \tilde{\rho}(\vec{r}_1 \mid \vec{r}_2)$

IDEAL GAS

$$\tilde{\rho}(\vec{r}_1 \mid \vec{r}_2) = \tilde{\rho}(\vec{r}_1) = \tilde{\rho}$$

HENCE

$$g^{(2)}(\vec{r}_1, \vec{r}_2) = \frac{\tilde{\rho}}{\tilde{\rho}} = 1$$

HARD SPHERES

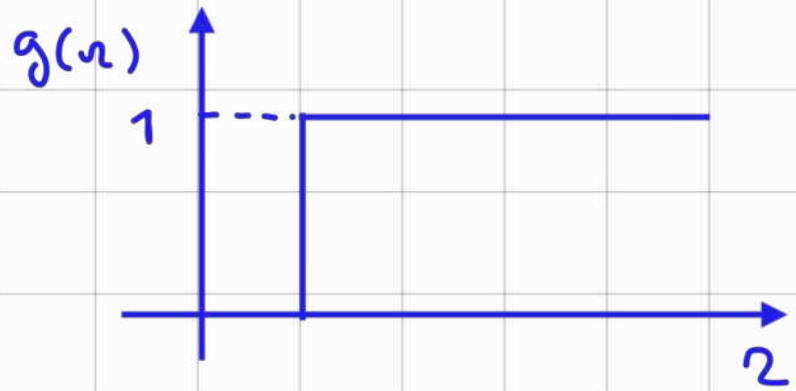
IF $\|\vec{r}_1 - \vec{r}_2\| < \sigma \Rightarrow \rho^{(2)}(\vec{r}_1 \mid \vec{r}_2) = 0$

AND $g^{(2)}(\vec{r}_1 \mid \vec{r}_2) = 0$

MOREOVER SYSTEM IS ISOTROPIC THEREFORE

$g^{(2)}(\vec{r}_1 \mid \vec{r}_2) = g^{(2)}(\|\vec{r}_{12}\|) = g^{(2)}(r)$

RADIAL DISTRIBUTION FUNCTION



AT VERY LOW DENSITIES

BUT AT HIGH DENSITIES ?

WE WILL SEE ...

TWO POINTS DENSITY

IF $\mathcal{N}(\vec{r}_1 \dots \vec{r}_N) = \dfrac{e^{-\beta U_N}}{Z}$

$\mathcal{N}(\vec{r}_1 \dots \vec{r}_N)\, d\vec{r}_1 \dots d\vec{r}_N$ IS THE PROBABILITY OF FINDING PARTICLES AT POSITIONS $\vec{r}_1, \dots \vec{r}_N$

HENCE THE PROBABILITY DENSITY TO FIND ONE PARTICLE IN $\vec{r}_1$ IS:

$$\tilde{\rho}(\vec{r}_1) = \int \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_2 \dots d\vec{r}_N$$

NOTE THAT WE ARE NOT INTEGRATING OVER $d\vec{r}_1$

AND THE (1-PARTICLE) DENSITY IS:

$$\rho(\vec{r}_1) = N \cdot \tilde{\rho}(\vec{r}_1) = N \cdot \int \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_2 \dots d\vec{r}_N$$

N: TOTAL NUMBER OF PARTICLES

NOTE THAT $\int \tilde{\rho}(\vec{r}_1)\, d\vec{r}_1 = 1$ AND THAT $N \tilde{\rho}(\vec{r}_1)\, d\vec{r} = \{$ NUMBER OF PARTICES IN $d\vec{r}_1 \}$

MOREOVER IF THE SYSTEM IS HOMOGENEOUS

$$\rho(\vec{r}_1) = \frac{N}{V}$$

SIMILARLY THE PROBABILITY DENSITY OF FINDING A PARTICLE IN $\vec{r}_1$ AND $\vec{r}_2$ IS

$$\tilde{\rho}(\vec{r}_1, \vec{r}_2) = \int \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_3 \dots d\vec{r}_N$$

NOTE THAT WE ARE NOT INTEGRATING OVER $\vec{r}_1$ AND $\vec{r}_2$

SO THAT

$\underbrace{N(N-1)}_{\text{TOTAL NUMBER OF POSSIBLE PAIRS}}\ \tilde{\rho}(\vec{r}_1,\vec{r}_2) = \{$ NUMBER OF PAIRS IN $d\vec{r}_1$ AND $d\vec{r}_2$ $\}$

AND WE DEFINE THE TWO PARTICLE DENSITY AS FOLLOWS

$$\rho^{(2)}(\vec{r}_1,\vec{r}_2) = N(N-1)\int \frac{e^{-\beta U_N}}{Z} d^{(N-2)}\vec{r}$$

IDEAL GAS

$$\rho^{(2)}(\vec{r}_1,\vec{r}_2) = N(N-1)\int \frac{e^{-\beta U_N}}{Z} d\vec{r}_3 \dots d\vec{r}_N$$

$U_N = 0$

$$= N(N-1)\int \frac{d\vec{r}_3 \dots d\vec{r}_N}{Z} = \overbrace{N(N-1)}^{\approx N^2} \frac{V^{N-2}}{V^N} =$$

SINCE $Z = \int e^{-\beta U_N} d\vec{r}_1 \dots d\vec{r}_N = V^N$

$$\approx \frac{N^2}{V^2} = \rho^2 \quad \text{HENCE}$$

$$g^{(2)}(\vec{r}_1,\vec{r}_2) = \frac{\rho^{(2)}(\vec{r}_1,\vec{r}_2)}{\rho^2} = \frac{\rho^2}{\rho^2} = 1$$

ALTERNATIVE WAYS OF CALCULATING 1-POINT AND 2-POINT DENSITIES

$U_N = U_N(\vec{r}, \vec{r}_2 \dots \vec{r}_N)$

$$\rho(\vec{r}) = \frac{N}{Z}\int e^{-\beta U_N} d^{(N-1)}\vec{r} =$$

NOW $U_N = U_N(\vec{r}_1 \dots \vec{r}_N)$

$$= \frac{N}{Z}\int e^{-\beta U_N} \delta(\vec{r}-\vec{r}_1)\, d^N\vec{r} =$$

INTEGRAL DOES NOT DEPEND ON PARTICLE INDEX

$$= \frac{1}{N}\sum_i^N \int e^{-\beta U_N} \delta(\vec{r}-\vec{r}_i)\, d^N\vec{r}$$

TAKE A SYSTEM OF 3 PARTICLES CONSIDER $\vec{r}_2$ AND $\vec{r}_3$ INSTEAD OF $\vec{r}_1$, THEN SWAP INDICES TO RECOVER CASE ABOVE ($\vec{r}_1$)

$$= \left\langle \sum_i^N \delta(\vec{r}-\vec{r}_i) \right\rangle$$

NOTE THAT $U_2(\vec{r}_{iT}) = U_2(\vec{r}_{Ti})$

WHERE $\langle \dots \rangle = \int \dots \frac{e^{-\beta U_N}}{Z} d^N\vec{r}$

SIMILARLY

$$\rho^{(2)}(\vec{r}', \vec{r}) = \left\langle \sum_{i,j}{}' \delta(\vec{r}' - \vec{r}_i)\, \delta(\vec{r} - \vec{r}_j) \right\rangle$$

($'$: $i \neq j$)

INDEED

$$\rho^{(2)}(\vec{r}', \vec{r}) = N(N-1) \int \frac{e^{-\beta U_N}}{Z} d^{(N-2)}\vec{r} =$$

($U_N = U_N(\vec{r}', \vec{r}, \vec{r}_3 \dots \vec{r}_N)$)

$$= N(N-1) \int \delta(\vec{r}' - \vec{r}_1)\, \delta(\vec{r} - \vec{r}_2) \frac{e^{-\beta U_N}}{Z} d\vec{r}_1 d\vec{r}_2 \cdot d^{(N-2)}\vec{r}$$

($U_N = U_N(\vec{r}_1 \dots \vec{r}_N)$)

TAKE AGAIN A SYSTEM OF 3 PARTICLES AND USE $(\vec{r}_2, \vec{r}_3)$ INSTEAD OF $(\vec{r}_1, \vec{r}_2)$

INTEGRAL DOES NOT DEPEND ON WHICH PAIR OF INDICES WE ARE CONSIDERING

$$= \sum_{i,j}{}' \int \delta(\vec{r}' - \vec{r}_i)\, \delta(\vec{r} - \vec{r}_j) \frac{e^{-\beta U_N}}{Z} d^N\vec{r} =$$

$$= \left\langle \sum_{ij}{}' \delta(\vec{r}' - \vec{r}_i)\, \delta(\vec{r} - \vec{r}_j) \right\rangle$$

$\rho^{(2)}(\vec{r}', \vec{r})$ IN THE HOMOGENEOUS CASE

LET'S CONSIDER

$$\left\langle \frac{1}{N\rho} \sum_{i,j}{}' \delta(\vec{r} - (\vec{r}_i - \vec{r}_j)) \right\rangle =$$

(FUNCTION OF $\vec{r}$ AND WE AVERAGE OVER $\vec{r}_1 \dots \vec{r}_N$)

$$= \left\langle \frac{1}{N\rho} \sum_{i,j}{}' \delta(\vec{r} - \vec{r}_i + \vec{r}_j) \right\rangle =$$

$$= \left\langle \frac{1}{N\rho} \int \sum_{i,j}{}' \delta(\vec{r} - \vec{r}_i + \vec{r}') \cdot \delta(\vec{r}' - \vec{r}_j)\, d\vec{r}' \right\rangle =$$

($\rho^{(2)}(\vec{r} + \vec{r}', \vec{r}')$)

$$= \frac{1}{N\rho} \int \left\langle \sum_{i,j}{}' \delta((\vec{r} + \vec{r}') - \vec{r}_i) \cdot \delta(\vec{r}' - \vec{r}_j) \right\rangle d\vec{r}' =$$

$$= \frac{1}{N\rho} \int \rho^{(2)}(\vec{r}+\vec{r}', \vec{r}')\, d\vec{r}' =$$

BUT THE SYSTEM IS HOMOGENEOUS

THEREFORE $\rho^{(2)}(\vec{r}+\vec{r}', \vec{r}') = \rho^{(1)}(\vec{r})$

I.E IT IS INDEPENDENT OF $\vec{r}'$

$$= \frac{1}{N\rho} \int \rho^{(1)}(\vec{r})\, d\vec{r}' = \frac{V}{N\rho}\, \rho^{(1)}(\vec{r})$$

$$= \frac{1}{\rho^2}\, \rho^{(1)}(\vec{r}) = \frac{g^{(2)}(\vec{r})\cdot \cancel{\rho^2}}{\cancel{\rho^2}} =$$

$$= g^{(2)}(\vec{r})$$

WE PROVED THAT

$$g^{(2)}(\vec{r}) = \frac{1}{N\rho} \left\langle \sum_{ij}{}' \delta(\vec{r} - (\vec{r}_i - \vec{r}_j)) \right\rangle$$

$$= \frac{1}{N\rho} \left\langle \sum_{ij}{}' \delta(\vec{r} - \vec{r}_{ij}) \right\rangle \qquad \vec{r}'_{ij} = \vec{r}_i - \vec{r}_j$$

AVERAGE VALUE OF $g(\vec{r})$ OVER A VOLUME ΔV AROUND $\vec{r}$

$$\bar{g}(\vec{r}) = \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} g(\vec{r}')\, d\vec{r}'$$

$\bar{g}(\vec{r})$: DEPENDENT ON CHOSEN $\Delta V_{\vec{r}}$ (IF $\Delta V_{\vec{r}}$ IS A CUBIC BOX, $\bar{g}$ DEPENDS ON ITS POSITION $\vec{r}$)

$\Delta V_{\vec{r}}$: THIS IS TO REMIND THAT VOLUME ΔV IS AROUND $\vec{r}$

IN A MC SIMULATION:

$$\langle O(\vec{r}^N) \rangle = \frac{1}{L} \sum_{k=1}^{L} O(r_k^N) = \frac{1}{L} \sum_{k=1}^{L} O_k$$

L: OR EVERY A CERTAIN AMOUNT OF STEPS

$\sum_{k=1}^{L}$: SUM OVER MC STEPS

HENCE CONSIDER THE FOLLOWING OBSERVABLE

$$\bar{g}(\vec{r}; \vec{r}_1 \ldots \vec{r}_N) \equiv \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{ij}{}' \delta(\vec{r}' - \vec{r}_{ij})\, d\vec{r}'$$

$\bar{g}(\vec{r})$ OF ONE CONFIGURATION (NO AVERAGE OVER $\frac{e^{-\beta U}}{Z}$)

ITS AVERAGE VALUE IS

$$\langle \bar{g} \rangle = \left\langle \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{iT}{}' \delta(\vec{r}' - \vec{r}_{iT})\, d\vec{r}' \right\rangle$$

$$= \int \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{iT}{}' \delta(\vec{r}' - \vec{r}_{iT})\, d\vec{r}' \frac{e^{-\beta U_N}}{Z}\, d\vec{r}_1 \dots d\vec{r}_N d\vec{r}$$

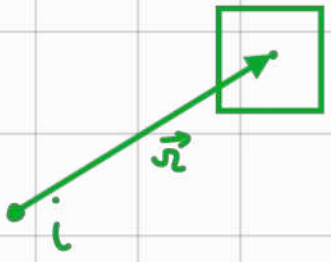
$$= \frac{1}{L} \sum_{k=1}^{L} \underbrace{\frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{i,T}{}' \delta(\vec{r}' - \vec{r}_{iT}^{(k)})\, d\vec{r}'}_{\bar{g}_k} = \frac{1}{L} \sum_{k}^{L} \bar{g}_k$$

$\bar{g}_k$ IS OUR OBSERVABLE (AS $\mathcal{O}_k$) CALCULATED AT k-th MC STEP

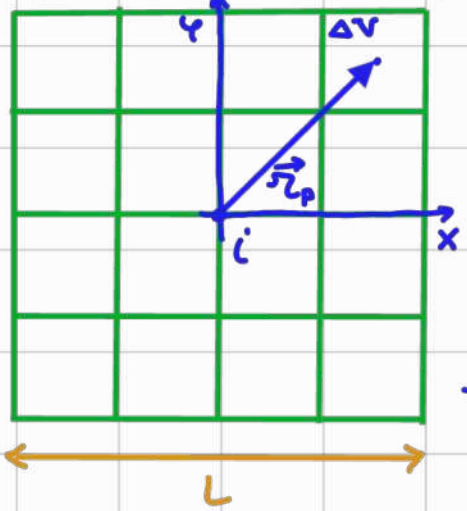
$$\bar{g}_k(\vec{r}) = \frac{1}{\Delta V} \int_{\Delta V_{\vec{r}}} \frac{1}{N\rho} \sum_{i,T}{}' \delta(\vec{r}' - \vec{r}_{iT}^{(k)})\, d\vec{r}'$$

AVERAGE OVER ALL PARTICLES

$$= \frac{1}{\Delta V \rho} \overbrace{\frac{1}{N} \sum_{i}}^{\text{AVERAGE OVER ALL PARTICLES}} \underbrace{\int_{\Delta V_{\vec{r}}} \sum_{T}{}' \delta(\vec{r}' - \vec{r}_{iT})\, d\vec{r}'}_{\#\text{ PARTICLES IN } \Delta V_{\vec{r}} \text{ A DISTANCE } \vec{r} \text{ FROM PARTICLE } i}$$



BOX CENTERED AT i OF VOLUME V DIVIDED INTO M SMALL BOXES OF VOLUME ΔV



$\vec{r}_p$ WITH $p = 1 \dots M$ IS THE CENTER OF p-th BOX OF VOLUME ΔV

$\vec{r}_p = \Delta r (k\hat{x} + \ell\hat{y} + m\hat{z})$

k, ℓ, m INTEGERS

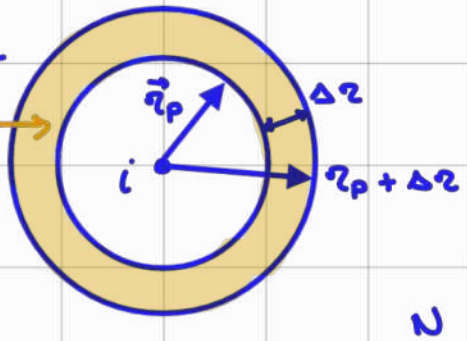
THUS OBTAINING

$\bar{g}_k(\vec{r}_p) \approx g_k(\vec{r}_p)$ IF ΔV SMALL ENOUGH

ISOTROPIC CASE

SPHERICAL SHELL OF THICKNESS Δr

$$\Delta V_p = 4\pi r_p^2 \Delta r$$

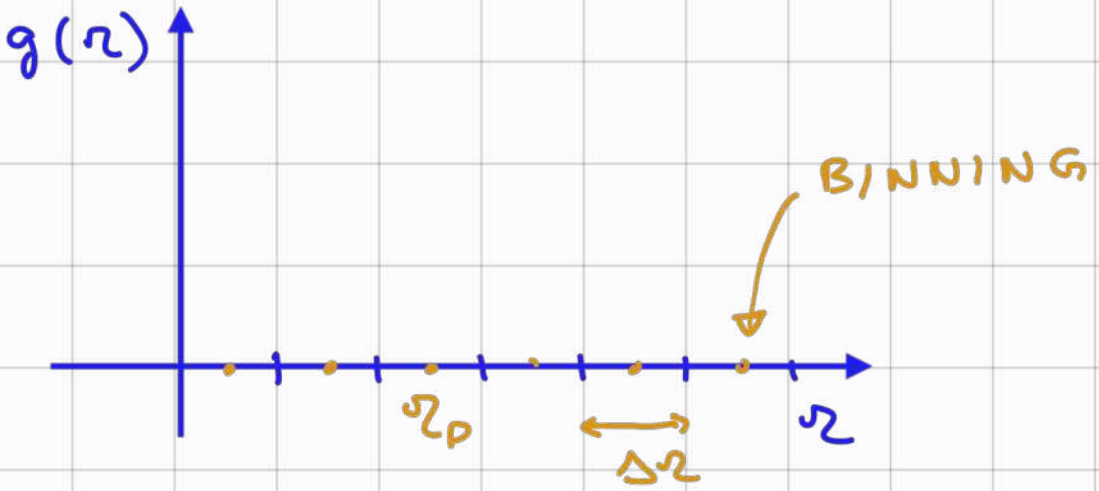


$$\overline{g}_k(r_p) = \underbrace{\frac{1}{4\pi r_p^2 \Delta r_p}}_{\text{AVERAGE NUMBER OF PARTICLES INSIDE } \Delta V_p} \frac{1}{N} \sum_i^N \int_{\Delta V_p} \overbrace{\sum_j{}' \delta(\vec{r} - \vec{r}_{ij})}^{\#\text{ PARTICLE INSIDE p-th SHELL}} d\vec{r} =$$

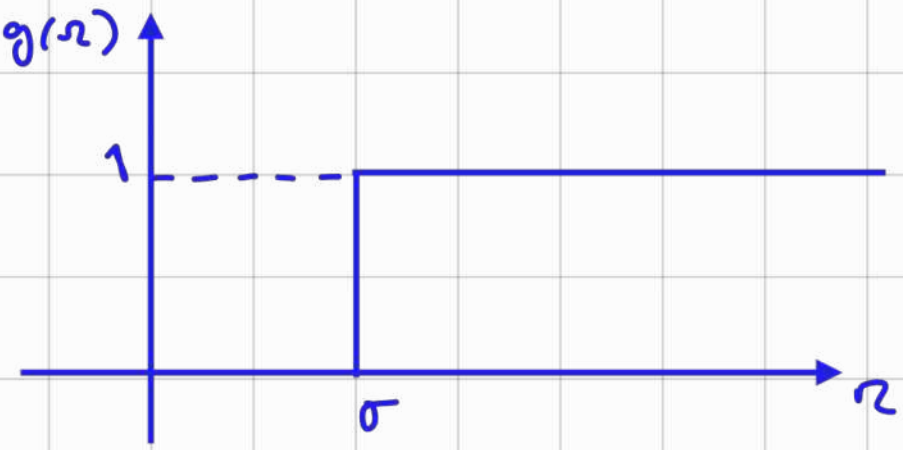
NOTE THAT THIS IS A SCALAR

$\overline{g}_k(r_p) \approx g_k(r_p)$ IS THE RADIAL DISTRIBUTION FUNCTION

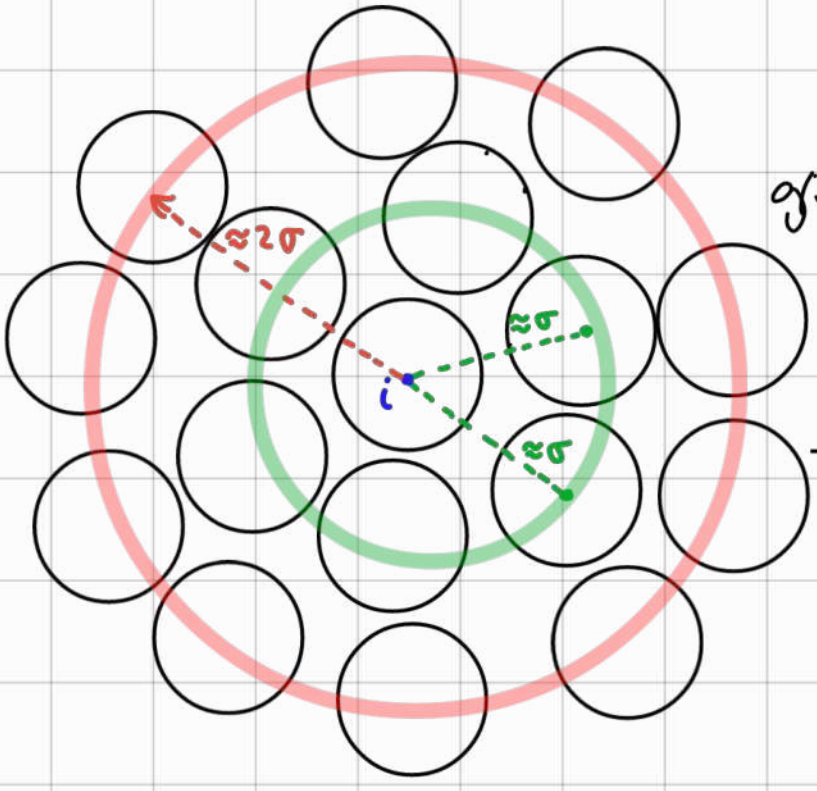
IF Δr IS SMALL ENOUGH

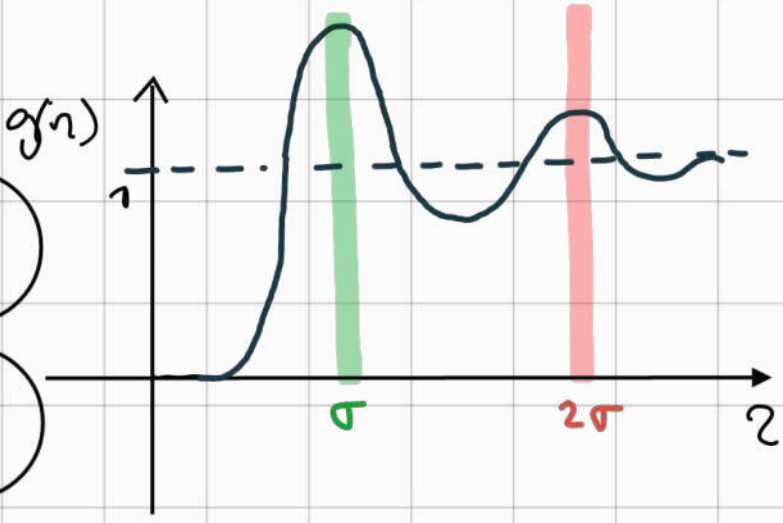


LOW CONCENTRATIONS



HIGH CONCENTRATION IN FLUID PHASE





CALCULATION OF ENERGY FROM g(r)

$$\langle U_N \rangle = U_{ex} = \int \frac{1}{2} \sum_{ij}{}' U_2(\vec{r}_{ij}) \frac{e^{-\beta U_N}}{Z} d^N\vec{r} =$$

$$= \frac{1}{2} N(N-1) \int U_2(\vec{r}_{12}) \frac{e^{-\beta U_N}}{Z} d\vec{r}_1 d\vec{r}_2 d^{(N-2)}\vec{r}$$

$$= \frac{1}{2} \int U_2(\vec{r}_{12}) \underbrace{\int N(N-1) \frac{e^{-\beta U_N}}{Z} d^{(N-2)}\vec{r}}_{\rho^{(2)}(\vec{r}_1, \vec{r}_2)} d\vec{r}_1 d\vec{r}_2$$

$$= \frac{1}{2} \int U_2(\vec{r}_{12}) \, \rho^{(2)}(\vec{r}_1, \vec{r}_2) \, d\vec{r}_1 d\vec{r}_2$$

BUT $\rho^{(2)}(\vec{r}_1, \vec{r}_2) = \rho^2 g(\vec{r}_{12})$ ← HOMOGENEOUS CASE

$$= \frac{\rho^2}{2} \int U_2(\vec{r}_{12}) \, g(\vec{r}_{12}) \, d\vec{r}_1 d\vec{r}_2$$

IF WE CHANGE VARIABLES

$(\vec{r}_1, \vec{r}_2) \to (\vec{r}_2, \vec{r})$ WHERE

$\vec{r} = \vec{r}_{12} = \vec{r}_1 - \vec{r}_2$

$$= \frac{\rho^2}{2} \int v_2(\vec{r})\, g(\vec{r})\, d\vec{r}\, d\vec{r}_2 =$$

$$= V \frac{\rho^2}{2} \int v_2(\vec{r})\, g(\vec{r})\, d\vec{r}$$

IF THE SYSTEM IS ISOTROPIC

$$= V \frac{\rho^2}{2} \int v_2(r)\, g(r)\, 4\pi r^2\, dr$$

↑ RADIAL DISTRIBUTION FUNCTION

TAIL CORRECTIONS

IN A SIMULATION $v_{ij}(r) \to v_{ij}^c(r)$

WHERE $v_{ij}^c = \begin{cases} v_{ij}(r) & r \le r_c \\ 0 & r > r_c \end{cases}$

HENCE

$$U_{ex} = \langle U_N \rangle = \frac{1}{2} V \rho^2 \int g(r)\, v_2(r)\, 4\pi r^2\, dr$$

U_{ex}^c THIS IS WHAT WE HAVE FROM SIMULATIONS, I.E. $\frac{1}{2}\sum_{i,j}{}' v_2^c(r_{ij})$

$$= \frac{1}{2} V \rho^2 \int_0^{r_c} g(r)\, v_2(r)\, 4\pi r^2\, dr +$$

$$+ \frac{1}{2} V \rho^2 \int_{r_c}^{\infty} g(r)\, v_2(r)\, 4\pi r^2\, dr$$

TAIL CORRECTION

Q. CAN WE ESTIMATE SOMEHOW THE TAIL CORRECTION?

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{\infty} g(r)\, u_2(r)\, 4\pi r^2\, dr$$

↑ TAIL CORRECTION

IF r_c IS BIG ENOUGH FOR $r \geq r_c$ $g(r) \approx 1$

HENCE

THIS INTEGRAL CAN BE CALCULATED ↓

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{\infty} u_2(r)\, 4\pi r^2\, dr$$

EXAMPLE LENNARD-JONES SYSTEM

$$u_2(r) = u_{LJ}(r) = A\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right]$$

SO THAT

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{\infty} 4\pi r^2 A\varepsilon\left[\left(\frac{\sigma}{r}\right)^{12} - \left(\frac{\sigma}{r}\right)^{6}\right] =$$

$$= V \frac{8\pi}{3} \varepsilon \rho^2 \sigma^3 \left[\frac{1}{3}\left(\frac{\sigma}{r_c}\right)^9 - \left(\frac{\sigma}{r_c}\right)^3\right]$$

WE CAN DO ALSO BETTER

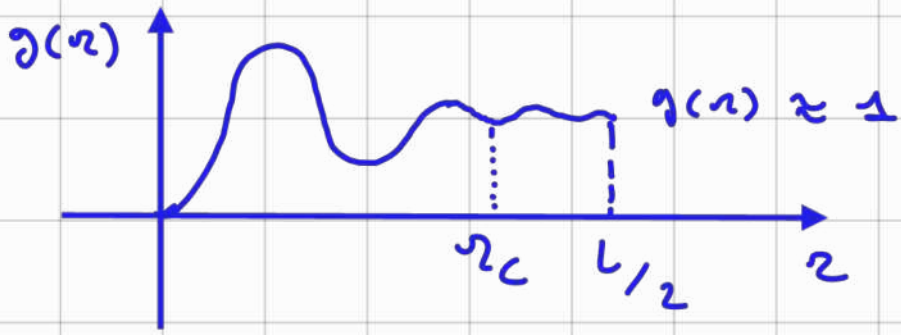
ENERGY FROM SIMULATION

$$U_{ex} = \underbrace{\frac{1}{2} V \rho^2 \int_0^{r_c} g(r)\, u_2(r)\, 4\pi r^2\, dr}_{\text{TAIL CONNECTION}} + \underbrace{\frac{1}{2} V \rho^2 \int_{r_c}^{L/2} g(r)\, u_2(r)\, 4\pi r^2\, dr}_{\approx 1} + \underbrace{\frac{1}{2} V \rho^2 \int_{L/2}^{+\infty} g(r)\, u_2(r)\, 4\pi r^2\, dr}_{\text{TAIL CORRECTION}} =$$

$$= U_{ex}^{C} + U_{ex}^{TAIL}$$

WHERE $g(r)$ CALCULATED FROM SIMULATIONS

$$U_{ex}^{TAIL} = \frac{1}{2} V \rho^2 \int_{r_c}^{L/2} g(r)\, u_L(r)\, 4\pi r^2\, dr + \frac{1}{2} V \rho^2 \int_{L/2}^{+\infty} u_L(r)\, 4\pi r^2\, dr$$



CALCULATION OF PRESSURE FROM $g(r)$

$$P = -\frac{\partial F}{\partial V} = k_B T \frac{\partial \ln Z}{\partial V} \qquad F = -k_B T \ln Q \quad \text{BUT}$$

$$Z = Z_{ID} \cdot Z_{ex} \qquad Z_{ID} = V^N \qquad Q = \frac{\Lambda^{-3N}}{N!} Z$$

$$Z_{ex} = \int e^{-\beta U_N(\vec{r}_1 \ldots \vec{r}_N)} d^N r$$

WHERE $\Lambda = \sqrt{\frac{2\pi\hbar^2}{m k_B T}}$

IT DOES NOT DEPEND ON V

SCALED COORDINATES:

$$\vec{r}_i = V^{1/3} \vec{s}_i$$

$$P = k_B T \frac{\partial}{\partial V} \left(\ln Z_{ID} + \ln Z_{ex} \right) = k_B T \frac{\partial \ln V^N}{\partial V} + k_B T \frac{\partial \ln Z_{ex}}{\partial V} = \underbrace{k_B T \rho}_{P_{ID}} + \underbrace{k_B T \frac{\partial \ln Z_{ex}}{\partial V}}_{P_{ex}}$$

$$P = P_{ID} + P_{ex}$$

$U_N = U_N(V^{1/3} s_1 \ldots V^{1/3} s_N)$

$$P_{ex} = k_B T \frac{1}{Z_{ex}} \frac{\partial Z_{ex}}{\partial V} = \frac{k_B T}{Z_{ex}} \frac{\partial}{\partial V} \left(\int e^{-\beta U_N} d^N s \right) =$$

$$= -\cancel{k_B T}\,\beta \int \frac{\partial U_N}{\partial V}\,\frac{e^{-\beta U_N} V^N}{Z_{ex}\, V^N}\, d\vec{s}^N = -\int \frac{\partial U_N}{\partial V}\,\frac{e^{-\beta U_N}}{Z}\, d\vec{r}^N =$$

$Z_{ex} V^N = Z$

$$= -\left\langle \frac{\partial U_N}{\partial V} \right\rangle$$

$$P = P_{ID} - \underbrace{\left\langle \frac{\partial U_N}{\partial V} \right\rangle}_{P_{EX}}$$

(*) NOTE THAT WE CAN BRING $\frac{\partial}{\partial V}$ INSIDE THE INTEGRAL SINCE THE INTEGRATION DOMAIN DOES NOT DEPEND ON V IF WE USE SCALE COORDS.

$$U_N = \frac{1}{2}\sum_{i,j}{}' U_2(\vec{r}_{ij}) \qquad V = L^3 \Rightarrow \frac{\partial}{\partial V} = \frac{1}{3L^2}\frac{\partial}{\partial L}$$

$$P_{EX} = -\left\langle \frac{1}{3L^2}\frac{\partial U_N}{\partial L} \right\rangle = -\left\langle \frac{1}{3L^2}\frac{\partial}{\partial L}\left(\frac{1}{2}\sum_{ij}{}' U_2(\vec{r}_{ij})\right)\right\rangle =$$

$\vec{s}_{ij} = \frac{\vec{r}_{ij}}{L}$

$$= -\left\langle \frac{1}{6L^2}\sum_{ij}{}' \underbrace{\frac{\partial U_2}{\partial \vec{r}_{ij}}}_{-\vec{f}_{ij}} \cdot \frac{\partial \vec{r}_{ij}}{\partial L} \right\rangle = \left\langle \frac{1}{6L^3}\sum_{ij}{}' \vec{f}_{ij}\cdot\vec{r}_{ij} \right\rangle =$$

NOTE THAT FOR A SYSTEM WITH PBC WE CAN RESTRICT THE CALCULATION TO "FIRST" BOX, SO THAT:

$$U_N = \frac{1}{2}\sum_{\vec{n}}\sum_{i,j} U_2(\vec{r}_i - \vec{r}_j + L\vec{n})$$

WHERE $\vec{n} = (n_x, n_y, n_z)$

WITH n_α INTEGERS

HENCE

$$\frac{\partial U_N}{\partial L} = \frac{1}{2}\sum_{ij}{}' \frac{\partial U_2(\vec{r}_i - (\vec{r}_j + L\vec{n}))}{\partial L} =$$

$$= \frac{1}{2}\sum_{ij}{}' \frac{\partial U_2}{\partial \vec{r}_{ij}} \cdot \frac{\partial(\vec{r}_i - (\vec{r}_j + L\vec{n}))}{\partial L} =$$

$$= \frac{r_i - (r_j + L\vec{n})}{L} = \frac{\vec{r}_{ij}}{L}$$

WHERE j RUNS OVER ALSO PERIODIC PARTICLES

(FORCE ACTING BETWEEN I AND J)

$$= \frac{1}{6V} \left\langle \sum_{ij}{}' \vec{f}_{ij} \cdot \vec{r}_{ij} \right\rangle$$

SIMILAR EXPRESSION FOR MOLECULAR PRESSURE

$$P = \rho k_B T + \underbrace{\frac{1}{6V} \left\langle \sum_{ij}{}' \vec{f}_{ij} \cdot \vec{r}_{ij} \right\rangle}_{P_{EX}}$$

WE CAN NOW WRITE P_{EX} IN TERMS OF $g(\vec{r})$

ALL TERMS OF $\sum_{ij}{}'$ ARE EQUAL

$$P_{EX} = \frac{N(N-1)}{6V} \int \frac{e^{-\beta U_N}}{Z} \vec{f}_{12} \cdot \vec{r}_{12} \, d\vec{r}_1 d\vec{r}_2 d^{(N-2)}\vec{r} =$$

$$= \frac{1}{6V} \int \rho^{(2)}(\vec{r}_1, \vec{r}_2) \, \vec{f}_{12} \cdot \vec{r}_{12} \, d\vec{r}_1 d\vec{r}$$

CHANGE OF VARIABLES

$(\vec{r}_1, \vec{r}_2) \rightarrow (\vec{r}_1, \vec{r})$ WHERE $\vec{r} = \vec{r}_1 - \vec{r}_2$

$$= \frac{1}{6V} \int \rho^{(2)}(\vec{r} + \vec{r}_2, \vec{r}_2) \, \vec{f}(\vec{r}) \cdot \vec{r} \, d\vec{r} \, d\vec{r}_2 =$$

IF SYSTEM IS HOMOGENEOUS

$\rho^{(1)}(\vec{r} + \vec{r}_2, \vec{r}_2) = \rho^{(2)}(\vec{r})$ HENCE

$$= \frac{1}{6} \int \rho^{(2)}(\vec{r}) \, \vec{f}(\vec{r}) \cdot \vec{r} \, d\vec{r}$$

BUT $\rho^{(2)}(\vec{r}) = \rho^2 g(\vec{r})$

$$= \frac{\rho^2}{6} \int \vec{f}(\vec{r}) \cdot \vec{r} \, g(\vec{r}) \, d\vec{r}$$

IF THE SYSTEM IS ISOTROPIC

$$P_{EX} = \frac{\rho^2}{6} \int f(r) \, g(r) \, 4\pi r^3 \, dr$$

TAIL CORRECTIONS

P_{SIM} FROM SIMULATIO

$$P_{EX} = \frac{\rho^2}{6}\int_0^{r_c} f(r)\, g(r)\, 4\pi r^3 dr + \underbrace{\frac{\rho^2}{6}\int_{r_c}^{+\infty} f(r)\, \cancel{g(r)}^{\approx 1}\, 4\pi r^3 dr}_{\text{FOR LJ WE CAN CALCULATE THIS ANALITICALLY}} = P_{TAIL}$$

(•) NOTE THAT

WE CAN CALCULATE THIS FROM SIMULATIONS

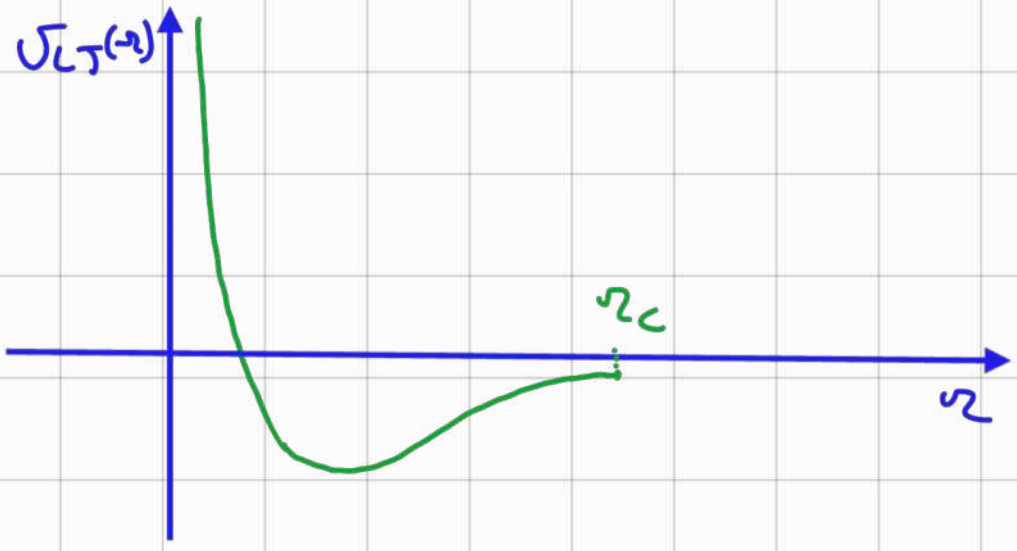
$$P_{SIM} = \frac{\rho^2}{6}\int_0^{r_c} f(r)\, g(r)\, 4\pi r^3 dr = \frac{1}{6V}\left\langle \sum_{ij}{}' \vec{F}_{ij}^{c} \cdot \vec{r}_{ij} \right\rangle$$

THE ABOVE TAIL CORRECTION APPLIES IF WE WANT TO COMPARE OUR RESULTS WITH A SYSTEM WITHOUT CUTOFF (•). E.G.

WE HAVE : $P = P_{SIM} + P_{TAIL}$

Q. BUT WHAT IS THE PRESSURE OF A SYSTEM WITH TRUNCATED INTERACTIONS?

FOR EXAMPLE CONSIDER A TRUNCATED LJ SYSTEM



AROUND r_c ONE HAS

$$v_{LT}(r) \approx \theta(r_c - r)\, v_{LT}(r_c)$$

WHERE

$$\theta(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}$$

$$f(r) = -\frac{dv_2}{dr} = v_{LT}(r_c)\,\delta(r - r_c)$$

SINCE

$$\frac{d}{dx}\theta(x) = \delta(x)$$

HENCE WE HAVE

$$P_{IMP} = \frac{1}{6}\rho^2 \int_{r_c-\varepsilon}^{\infty} v_{LT}(r_c)\,\delta(r - r_c)\,4\pi r^3\,dr =$$

$$= \frac{1}{6}\rho^2\, v_{LT}(r_c)\,4\pi r_c^3$$

"IMPULSIVE" CORRECTION TO PRESSURE

STATIC STRUCTURE FACTOR

1 POINT DENSITY

$$\rho(\vec{r}) = \left\langle \sum_i \delta(\vec{r} - \vec{r}_i) \right\rangle$$

THEN

$$\langle \rho(\vec{k}) \rangle = \underbrace{\int \rho(\vec{r}) e^{-i\vec{k}\cdot\vec{r}} \, d\vec{r}}_{\text{FOURIER TRANSFORM}} = \left\langle \underbrace{\sum_i e^{-i\vec{k}\cdot\vec{r}_i}}_{\rho(\vec{k})} \right\rangle$$

↑ ENSEMBLE AVERAGE

WHERE $\rho(\vec{k}) = \sum_i e^{-i\vec{k}\cdot\vec{r}_i}$

WE DEFINE

$$S(\vec{k}) \equiv \frac{1}{N} \langle \rho(\vec{k}) \rho^*(\vec{k}) \rangle$$

↑ ENSEMBLE AVERAGE

← $S(\vec{k})$ CAN BE MEASURED EXPERIMENTALLY THROUGH SCATTERING EXPERIMENTS

BUT

$$S(\vec{k}) = \frac{1}{N} \left\langle \sum_{i,j} e^{-ik(\vec{r}_i - \vec{r}_j)} \right\rangle = 1 + \frac{1}{N} \left\langle \sum_{i,j}' e^{-i\vec{k}\cdot(\vec{r}_i - \vec{r}_j)} \right\rangle =$$

($\sum'$: $i \neq j$)

$$= 1 + \frac{1}{N} \left\langle \int \sum_{i,j}' e^{-ik\cdot(r \rightleftarrows r')} \delta(\vec{r} - \vec{r}_i)\, \delta(\vec{r}' - \vec{r}_j) \, d\vec{r}\, d\vec{r}' \right\rangle =$$

$$= 1 + \frac{1}{N} \int d\vec{r}\, d\vec{r}' \, e^{-i\vec{k}\cdot(\vec{r} - \vec{r}')} \underbrace{\left\langle \sum_{i,j}' \delta(\vec{r} - \vec{r}_i)\, \delta(\vec{r} - \vec{r}_j) \right\rangle}_{\rho^{(2)}(\vec{r}, \vec{r}')} =$$

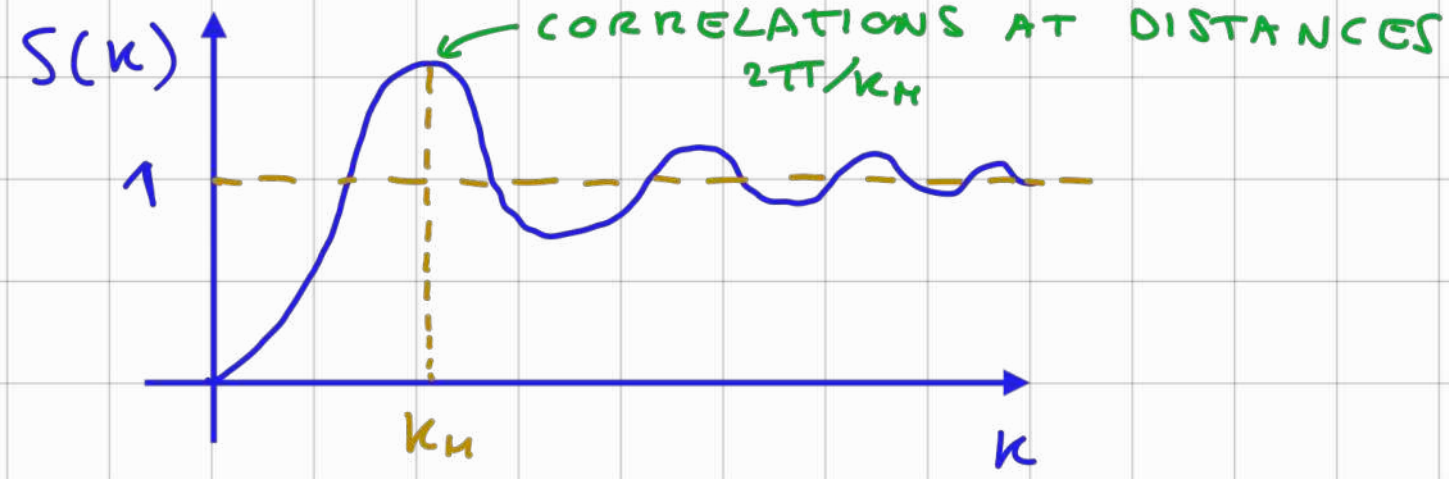
$$= 1 + \frac{1}{N} \int d\vec{r}\, d\vec{r}' \, e^{-ik(\vec{r} - \vec{r}')} \underbrace{\rho^2 g^{(2)}(\vec{r}, \vec{r}')}_{\rho^{(2)}(\vec{r}, \vec{r}')} =$$

ASSUMING THE SYSTEM HOMOGENEOUS

$$= 1 + \frac{\rho^2}{N} V \int d\vec{r} \, e^{-i\vec{k}\cdot\vec{r}} \, g(\vec{r}) =$$

$$= 1 + \rho \underbrace{\int d\vec{r} \, e^{-i\vec{k}\cdot\vec{r}} \, g(\vec{r})}_{\text{FOURIER TRANSFORM OF } g(\vec{r})}$$

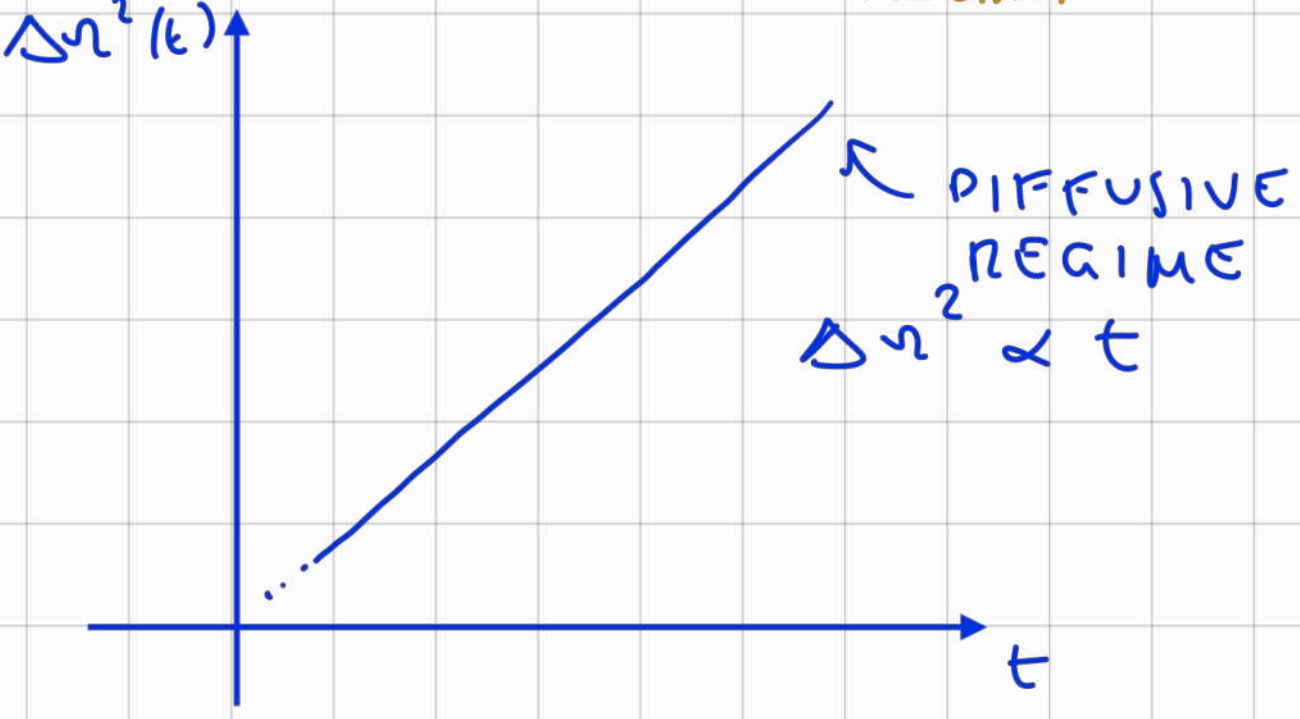
IN THE ISOTROPIC CASE



MEAN SQUARED DISPLACEMENT

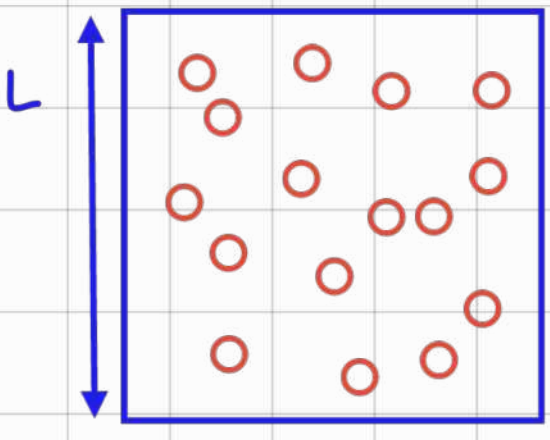
$$\Delta r^2(t) \equiv \frac{1}{N} \sum_{i=1}^{N} \| \vec{r}_i(t) - \vec{r}_i(0) \|^2$$

- $\Delta r^2(t)$: FUNCTION OF t
- $\vec{r}_i(t)$: PASSO MC
- $\vec{r}_i(0)$: POSIZIONI INIZIALI



DIFFUSIVE REGIME

$\Delta r^2 \propto t$



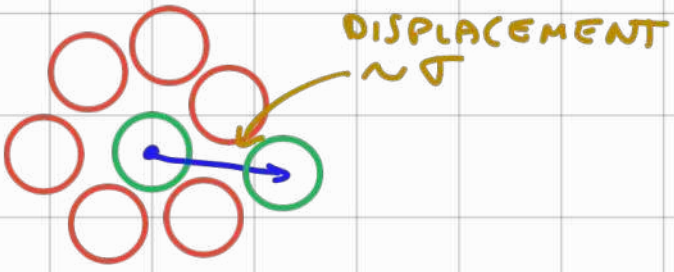
$\sqrt{\Delta r^2} \gtrsim L$

THE SYSTEM CAN BE CONSIDERED EQUILIBRATED SINCE ALL PARTICLES HAVE MOVED ENOUGH I.E. HAVE FULLY EXPLORED THE CONFIGURATION SPACE

WE COULD ALSO USE A LESS STRINGENT CONDITION, I.E.

TYPICAL PARTICLE SIZE

$$\sqrt{\Delta r^2} \gtrsim \sigma$$



MAIN CONTRIBUTION TO SPATIAL CORRELATIONS COMES FROM FIRST PEAK OF $S(R)$ WHICH ROUGHLY CORRESPONDS TO A DISTANCE EQUAL TO σ